Generative Linguistics Meets Normative Inferentialism*

David Pereplyotchik
Kent State University

Forthcoming in the Croatian Journal of Philosophy
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Abstract:

Squarely in the Chomskyan tradition, Paul Pietroski's recent book, *Conjoining Meanings*, offers an approach to natural-language semantics that rejects foundational assumptions widely held amongst philosophers and linguists. In particular, he argues against the view that meanings are, or at least determine, *extensions (*truth conditions, satisfaction conditions, and denotation/reference). Having arrived at the same conclusion by way of Brandom's deflationist account of truth and reference, I glimpse the possibility of a fruitful merger of ideas. In the present essay, I outline a strategy for integrating the generative linguist's empirical insights about human psychology with Brandom's pragmatist approach to language. I'll argue that both have important contributions to make to our overall understanding of language, and that the differences between them almost all reduce to a cluster of interrelated verbal differences. Contrary to first appearances, there are actually very few points of substantive disagreement between them. The residual differences are, however, stubborn. I end by raising a question about how to square Pietroski's commitment to predicativism with Brandom's argument that a predicativist language is in principle incapable of expressing ordinary conditionals.

Keywords

generative linguistics, anti-extensionalism, normativity, inferentialism, predicativism, public language, communication

* My deepest thanks to Daniel Harris, Ryan DeChant, Elmar Unnsteinsson, Eliot Michaelson, and Jessica Lynne Cooperrider for helpful advice on earlier drafts of this paper; thanks also to Nate Charlow, Matt Moss, and other participants of the 2018 Ontario Meaning Workshop, for stimulating exchanges on related topics. That said, if some of what I claim here is false, then these people are all partly to blame. Ditto for cases where I say something awkwarder than usual; they should of caught that. My sincere gratitude, lastly, to all of the participants at the two symposia at which I was fortunate to present earlier versions of these ideas, including the 2018 Croatian Meaning Workshop. From the latter group, I owe special thanks to Paul Pietroski (obviously), Michael Devitt, John Collins, Anna Drożdżowicz, and, most of all, to Dunja Jutronić—not only for inviting me to participate in the workshop, but for her saintly and near-inexhaustible patience with my obnoxious delays in submitting the present work to the Journal. Not to mention the outrageous word-count. I beg the reader's forgiveness for the latter, as well.

Introduction

I take it that the correct approach to natural-language syntax is the one that Noam Chomsky outlined as early as the 1950s and, along with many others, has continually refined over the past seven decades. The ongoing research program of generative linguistics that his syntactic theorizing inspired has, in the fullness of time, yielded a diversity of impressive results. These include exciting and previously unimaginable empirical discoveries about the human capacity for language, both in broad scope—e.g., recursive generability and the principles-and-parameters model—and at the level of fine-structure (e.g., traces, parasitic gaps, etc.). However, as a theorist interested not only in syntax but also in semantics, I find myself in a difficult and somewhat awkward position.

Not to complain, but, you see, I happen to have learned my semantics from the work of Robert Brandom, and it's safe to say that I've drunk the Kool-Aid that he served up in his magnum opus, *Making It Explicit*. Having thus bought into *both* Chomsky's generative grammar *and* Brandom's normative inferentialism, I now find myself facing the daunting challenge of bridging the apparent chasm between the two. It may be that I'm utterly alone in this quandary, but in the course of the present discussion, I hope—perhaps somewhat perversely—to draw others into it as well.

There may never be an academic conference addressing the common themes and shared commitments of generative linguistics and normative inferentialism. For all that, the differences between them are, I believe, more boring—i.e., verbal or sociological—than is widely assumed. In what follows, I'll argue that, contrary to first appearances, there are actually very *few* points of substantive disagreement between them. The residual differences are stubborn, to be sure, but this can only be appreciated after a suitably wide collection of background agreements is put into place. I devote the first half of the present discussion to this latter task.

Squarely in the Chomskyan tradition, Paul Pietroski's recent book, *Conjoining Meanings*, offers an approach to natural-language semantics that rejects foundational assumptions that are widely-held amongst philosophers and linguists. In particular, he argues against the view that meanings are, or at least determine, *extensions*. The latter include such familiar semantic properties as truth conditions, satisfaction conditions, and denotation/reference. Having arrived at the same conclusion by way of Brandom's deflationist account of truth (§1.3), I began to glimpse the possibility of a fruitful merger of ideas. In what follows, I take a first pass at integrating the generative linguist's empirical insights about human psychology with the broadly pragmatist framework about mind and language that Brandom has developed over the course of his career.[1] I'll argue that both have important contributions to make to our overall understanding of language. The easy part is spelling out what; the harder part is assessing the residual disagreements.

Here's the overall plan:

In §1, I survey a range of core commitments that jointly constitute Brandom's philosophical project—most centrally, his normative pragmatics, inferentialist semantics, and substitutional syntax. Along the way, I note his intellectual debts to David Lewis, including the large-scale

[1] The picture I sketch in what follows is drawn mostly from the material in Brandom (1994, 2000, and 2008). In what follows, I'll occasionally abbreviate these to *MIE* (*Making it Explicit*), *AR (Articulating Reasons)*, and *BSD (Between Saying and Doing*, 2008*)*, respectively. The final sections of this essay also make heavy use of the material in "Why Philosophy Has Failed Cognitive Science" (in Brandom, 2009).

explanatory goals that animate Brandom's inquiry. These, I later argue, are in many ways orthogonal to Pietroski's concerns. The latter claim figures in my broader argument that the two approaches can be fruitfully combined, potential protests from both sides notwithstanding.

In §2, I outline Pietroski's position, focusing on his explanatory aims, his empirical methodology, and the substance of his proposal for a theory of human semantic competence. I take up Pietroski's arguments against Lewis's approach to natural language, with the aim of showing that Brandom's theoretical goals differ sufficiently from Lewis's to inoculate him against Pietroski's criticisms. Turning to Pietroski's discussions of Frege, I point out that his work in cognitive science undercuts some of Brandom's claims in "Why Philosophy Has Failed Cognitive Science". As we'll see, far from ignoring Frege, Pietroski incorporates many of his insights into an empirical account of psychological processes. Nevertheless, some of the Fregean lessons that Brandom emphasizes do *not* seem to have moved Pietroski. In later sections, I explore some possible reasons for this.

I devote §3 to a survey of the core commitments that Pietroski and Brandom have in common. As already noted, both reject truth-conditional semantics and seek to develop alternative frameworks for semantic theorizing. Similarly, I'll point out, the alternatives they propose can both be seen as taking referential *purport* to be the distinctive feature of language. This contrasts with received views in (meta)semantics that focus, in the first instance, on referential *success*, leaving reference *failure* (e.g., empty names) for special treatment, as a blessedly rare "defective" case. This is related, I suspect, to the further convergence between the Brandom and the Pietroski on the proper treatment of the distinction between *de dicto* and *de re* constructions. Last, but no less important, is their common rejection of the idea that communication requires an *identity* between the meanings expressed by a speaker and those grasped by the hearer. I'll note that this shared commitment undercuts some of the main arguments against meaning holism—an inherent feature of Brandom's inferentialist account.

A discussion of the differences between Pietroski and Brandom occupies the remainder of the essay. At first glance, the contrast between them seems as vivid as any in the field. Brandom's project is explicitly *normative*, pursued largely from the armchair, and aims to provide an account of concept users *as such*—not just humans, but *any* conceivable concept-mongering creature (or artificial system). By contrast, Pietroski's project is avowedly *descriptive*, constrained by *empirical* data, and aims to provide an account of *actual humans*—particularly, the "Slangs" that they naturally produce and consume. (Pietroski uses 'Slang' as a catch-all term for the natural languages that children acquire.) These differences ramify quickly. For instance, Brandom's focus on social practices of communication seems to be at odds with Pietroski's "individualistic" methodology, which is characteristic of generativist views more broadly (Chomsky, 2000; Collins, 2012). This contrast seems particularly sharp in light of Brandom's commitment to the existence of (something like) public languages—at least in the sense of productive and flexible norms governing the communal practice of "giving and asking for reasons" (GOGAR). All of these issues are discussed in §4, with the aim of gradually blunting the force of what initially appear to be quite sharp differences.

If one insists on seeing Brandom's and Pietroski's inquiries as targeting a common subject matter, then one is sure to view the differences between them as substantive theoretical *disagreements*. They do, after all, use the same term, 'language', which at least *suggests* that they're talking about one and the same phenomenon. But this way of viewing the situation is optional, at best. We've learned by now to stop assuming that theorists are targeting the same phenomenon simply on account of their using homophonous terms. Two theorists can press the same bit of folk vocabulary, e.g., 'meaning' or 'concept', into more weighty theoretical labor in quite different ways. That being so, one can just as well see the inferentialist and the generativist as addressing *different* (though undoubtedly related) topics—each providing insights about his chosen domain of inquiry, and leaving the rest of us to wonder how those insights might be integrated, or at least brought to bear on one another. This is the strategy I'll recommend throughout the present discussion.

In arguing that the two are, at the very least, logically *compatible*, one owes an account—or, at a minimum, a broad *sketch* of an account—of the theoretical *relation* between them. The proposal that I'll develop is that Brandom's explanatory ambitions differ from Pietroski's in precisely the ways that are paradigmatic of "*inter-level* theoretical relations". My suggestion is that Brandom is attempting to furnish a *high-level description* of a quite general phenomenon—*language use, as such*. This presupposes that lower-level implementations of the relevant generalizations can vary widely.[2] Pietroski's theoretical aims are, though certainly more exciting from an empirical standpoint, a good bit narrower than Brandom's, in that they deal exclusively with the human case. As with any lower-level account of a more general phenomenon, Pietroski's view is compatible, in principle with any number of higher-level descriptions of language. Thus, while it's incorrect—or, at any rate, misleading—to say that the two accounts are strictly orthogonal to one another, the fact is that each places very few constraints on the other. It *should* be surprising, then, when we find substantive points of contact between them, whether these be points of contention or convergence.

Still, even if my suggestion is right that the two frameworks are more compatible than might initially appear, we must face up to the residual differences that credibly threaten my reconciling project. The most stubborn of these, which I accordingly leave to the very end, has to do with Pietroski's arguments for a specific version of *predicativism*—roughly, the view that all subsentential entities are best treated as predicates, not (say) singular terms (§2.4). To be sure, Pietroski's commitment to this view is not a central aspect of the overall generativist enterprise. Rather, it's a tendentious empirical hypothesis, for which he offers correspondingly forceful arguments. That being so, if it were to turn out that the hypothesis is false, generative linguistics would go on without a hitch. Still, I focus on this issue because it raises much larger questions about how to treat subsentential entities, not just at the level of semantics, but also at the level of *syntax*.

The notion of syntax that Brandom employs is *substitutional*—not in the sense of "substitutional quantification" (though he endorses that too, on independent grounds), but, rather, in the sense that he takes *sentences,* i.e., expressions of *full thoughts*, to be the primary vehicles of meaning. On his view, subsentential items (words, morphemes, etc.) are the products of taking a "substitutional scalpel" to the antecedently-interpreted *sentential* unities. We'll take a first stab at unpacking the scalpel metaphor in §1.5, and then come back to it in more detail in §4.3. While this substitutional approach *may* have a home in a Fregean semantics—Pietroski's powerful arguments notwithstanding—there is no *obvious* sense in which it can be legitimately applied to the syntax of human languages.

The trick that I pull repeatedly throughout the second half of this essay—namely, that of relegating the two inquiries to distinct theoretical "levels"—doesn't get much of a grip here. For, it's a requirement of any such picture that an account pitched at the (relatively) higher level of analysis should be *compatible*, at least in *core* respects, with any lower-level account of the

[2] Indeed, as Fodor (1975) points out, without substantive constraints from the higher-level account, such "realization bases" might differ *indefinitely*.

"realization base". But Brandom's substitutional syntax seems, even upon close scrutiny, to be not only *different* from generative grammar, but decidedly *at odds* with it. I strongly suspect that this different comes down to a methodological conflict between the generativist's "bottom-up" treatment of subsentential items and the inferentialist's "top-down" alternative.[3] In §4, I discuss and evaluate some ways of viewing this disagreement, arriving ultimately at the following bittersweet conclusions.

Despite strong grounds for optimism about the possibility of integrating normative inferentialism with the up-and-running research program of generative linguistics, it must be admitted that rendering Brandom's substitutional approach to syntax compatible with the going theories in generative grammar (e.g., Chomsky's minimalist program) presents an obstinate challenge. Though I can think of no reason that the challenge is insuperable *in principle*, it's nevertheless the case that I am not, at present, equipped to meet it myself. Perhaps others can do better in this regard—a task I invite, encourage, and exhort philosophers of language to undertake, on the strength of the positive arguments adduced here.

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§1. Brandom: From Normative Pragmatics to Inferentialist Semantics (and Back)

Introduction

Robert Brandom's philosophical project is grand in both scope and ambition. The resulting theoretical framework has a number of moving parts, to put it mildly. In this section, I'll lay out what I take to be the central commitments of his "normative inferentialism", with particular focus on those that pertain to the broader goals of this discussion—i.e., the proposal to integrate the generativist and inferentialist research programs.

Although this section is intended to be largely exegetical, postponing critical evaluation to §4, I should emphasize that it wouldn't matter to me very much if Brandom wouldn't put things *quite* the way that I do below. While his account is by far the most well-worked out version of normative inferentialism, and hence immensely useful as a guide in this area of inquiry, my intent is not so much to capture every nook and cranny of one particular theorist's gargantuan philosophical system. Rather, the goal of this section is to present and motivate an account of language that is attractive enough to warrant comparison with Pietroski's independently attractive proposals about natural-language semantics (§2). Ultimately, it's only by comparing them that we can put ourselves in a position to clearly assess the merits of either, let alone to contemplate their integration.

§1.1 Methodology and Explanatory Aims

The central questions, for Brandom, are about what *constitutes* language use—not by humans, necessarily, but by any natural creature or artificial system to which attributions of a linguistic competence are warranted. Given that normal adult humans have mastery of at least one natural language, Brandom's account will, in *some* sense, apply to us as well—though perhaps only as a particular instance of a much more general phenomenon. Still, the inquiry he undertakes is not a straightforwardly psychological one; nor is it pitched as a historical or

[3] See Collins (2012) for a detailed and rewarding discussion of this issue. The contrast that Collins draws between sentence-first and word-first approaches (my terms, not his) serve, if anything, to *sharpen* the contrast that I'm worried about here. My hunch is that coming to grips with Collins' conclusions in that work will be crucial to resolving the issues I raise in §4.3.

anthropological hypothesis. Rather, the idea is that, armed with a philosophically sophisticated and conceptually articulated account of "what the trick *is*"—where "the trick" is language use, broadly construed—an empirical scientist (a linguist, psychologist, or artificial intelligence researcher) can ask more detailed questions about "how the trick *happens to be done*" by one or another creature or artifact.

This latter kind of research is bound to yield an increasingly refined picture of some particular *type* of linguistic competence—paradigmatically, the human type (though one often hears of impressive progress in interpreting the languages of other social creatures, e.g., prairie dogs and dolphins). Thus, although the account is intended to apply far beyond the human case—to aliens, robots, and other terrestrial animals—there is no commitment on Brandom's part to the effect that empirical findings can have no bearing *whatsoever* on his philosophical claims. Nor does he hold that legitimate empirical inquiry can take place only *after* a credible philosophical account has been supplied. This is just one sense in which normative inferentialism and empirical science, including generative linguistics, are not in competition.

One might wonder, at this point, how far removed Brandom's project is from empirical considerations. Perhaps *too* far? True, many of the thinkers with whom his work most directly engages did *not* conceive of their inquiries on the model of empirical theorizing. In fact, the three philosophers who figure most centrally in *MIE*—namely, Kant, Frege, and Wittgenstein—all famously made a point of *distancing* themselves from natural science. But we have to keep in mind, here as elsewhere, that Brandom's theoretical framework is "a house with many mansions". Thus, we need not take a one-size-fits-all approach to this question; indeed, there are good reasons *not* to.

In providing detailed analyses of linguistic constructions in distinctively human languages—*de dicto/re* reports, truth-talk, (in)definite descriptions, indexicals, deixis, anaphora, modals, quantification, predicates, and singular terms—Brandom relies on exactly the same stock of empirical considerations that one finds in standard semantics texts (e.g., Heim and Kratzer, 1998). Such data are often cast in terms of intuitive judgments concerning the *truth*-values or *truth*-conditions of a target sentence, often in the context of auxiliary assumptions that are supplied by an accompanying vignette. But these very same data can equally well be recast as competent speakers' judgements concerning (not truth but) *inferential proprieties*—e.g., what one is required or permitted to infer on the basis of the assumptions supplied (in combination with all prior background information that is assumed to be common-knowledge).[4]

A similar treatment can be given of metalinguistic intutions/judgments concerning minimal pairs and sentential ambiguity. Two sentences differ in meaning, on the inferentialist account, just in case they differ in respect of the inferential proprieties that govern their use. Likewise, a sentence is ambiguous just in case it is capable of playing two distinct/incompatible inferential roles. And, though he doesn't, to my knowledge, ever discuss the phenomenon of polysemy, I presume that Brandom would treat is as a case of *overlapping* inferential properties—perhaps ones that meet some further normative or inferential conditions.

The fact that Brandom often appeals to precisely such data in developing his inferentialist semantics suggests that at least *this* aspect of his view is firmly grounded in empirical fact. But, as I'll emphasize later, Brandom's use of these data is not *empirical* but, rather, *illustrative*. He is, in other words, using examples from English—indeed, almost *exclusively* English, in contrast

[4] My thanks to Eliot Michaelson and Daniel Harris for helping me to see the points in this paragraph more clearly.

with the generativist's cross-linguistic methodology—as *case studies* that, according to him, exemplify *more general* phenomena. Thus, the English-language examples that he occasionally provides—a bit too rarely, one might lament—serve not as *empirical data* for a scientific or naturalistic hypothesis. Rather, the most charitable reading of his appeals to such examples casts them as attempts to help us to get a conceptual grip on features of language(s) *as such*.

Still, it must be admitted that other aspects of Brandom's overall picture are far less tethered to the facts on the ground. Presently, we'll see that his normative pragmatics—to which his downstream proposals, *including inferentialism*, are conceptually subordinate—makes only very minimal empirical assumptions. For instance, it takes for granted that complex social creatures came into being *somehow or other*—e.g., via evolution by natural selection, or by deliberate engineering, as with AI. Brandom likewise presumes that the behavioral control systems of such creatures/artifacts—brains, motherboards, or what have you—work *somehow or other*; else they couldn't perform the behaviors that institute social norms, given reasonable naturalistic constraints. Still, putting aside such near-vacuous assumptions, this aspect of Brandom's project can fairly be described as an armchair enterprise.[5] Whether that's something to hold *against* it—e.g., on the basis of one or another naturalist scruple—is something we can only ascertain only after we've surveyed the details of his overall proposal. To these we now turn.

§1.2 Normative Pragmatics

Brandom begins by situating all linguistic practices in the wider realm of activities that are, in some sense, *rule-governed*. The notion of a rule plainly stands in need of careful articulation. Given our philosophical history, two options immediately suggest themselves—what Brandom calls "regulism" and "regularism". According to the *regulist*, rule-following is a matter of obeying rules that one can *explicitly* formulate or comprehend. By contrast, the *regularist* holds that rule-following is a matter of being *disposed to behave* in a way that *accords* with one or another empirical *regularity*. Brandom rejects both of these alternatives, though, as we'll see, his own account is an attempt to split the difference between them.

Regulism is vitiated, he argues, by the fact that obeying an explicit rule—e.g., a dictionary definition or an academic/prescriptive convention of grammar or style—requires first *interpreting* the rule. This, in turn, requires *deploying concepts*—a version of precisely the phenomenon that we're seeking to explain. The regularist option, he contends, faces a distinct challenge. A pattern of behavior—whether finite or infinite, actual or potential—can be either a *successful* or a *defective* case of following a given rule. So, with regard to *any* pattern of performance, the question always stands: has the rule in question been followed *correctly*? Assessments of correctness are, on Brandom's view, inherently *normative*. Yet, what the regularist offers is a purely *descriptive* account, couched in the language of cognitive or behavioral dispositions (and perhaps other, related alethic modal notions; see §3.1).

Brandom's positive proposal insists on the normativity of assessment, characteristic of the regulist view, but jettisons the requirement that the rules in question be manifested *explicitly* from the very outset. Though rules of practice can *eventually* come to be articulated—i.e., "made explicit"—by a community of concept-using creatures, such rules are, in the first instance, *implicit* in the social practices of the creatures in question.

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[5] Notably, a former student of Brandom's, John MacFarlane, has programmed a version of "the game of giving and asking for reasons" (GOGAR) for the popular game, The Sims. https://www.johnmacfarlane.net/gogar.html

Why social? Why not, instead, the practices of an individual? Put crudely, the reason is that isolated individuals cannot, in principle, serve as a normative check on their own judgment and behavior. In order for a creature to be so much as *subject* to normative assessment, its behavior must take place in a context where other creatures respond to it in ways that signal social (dis)approval. How numerous and long-lasting must the social relations be in order to institute a social practice, properly so called? Brandom's answer, which will become relevant later in the discussion (§4.1), may be somewhat surprising. Terms like 'social' and 'communal' bring to mind a relatively large group of creatures. But, when he presses these folk terms into theoretical service, Brandom's official view is far less committal than all that. All it really requires is a *dyadic* "I-thou" relation—i.e., a case of mutual recognition, in respect of authority and responsibility, on the part of at least *two* creatures/systems. Such relations of mutual recognition can be merely *implicit* in the 2+ creatures' overt practices toward one another—e.g., one or another type of *social sanction*.

In the most basic case, social sanctions come down to either naked violence or the provision of necessities—beatings and feedings. But once a social practice becomes sufficiently complex, it comes to include not only such "external" sanctions, but also "internal" ones—e.g., initially, the granting of privileges and later the exchange of *tokens* of privilege. (We are invited here to imagine a special fruit that permits its bearer to enter a particular territory without being attacked by the creatures who guard it.) With each additional layer of interwoven internal sanctions, the community becomes increasingly ripe for instituting not merely norms of practical action, broadly construed, but the more specifically *linguistic* norms of assertion and demand. I include the latter here, not because Brandom ever treats it in detail, but on account of his frequent invocation of the trope "the game of giving *and asking for* reasons" (my emphasis, to be sure). While the "asking" part seems to deserve equal attention, Brandom takes the norms governing *assertion* to be the "downtown" of language—a backhanded adaptation of Wittgenstein's metaphor of language as a city with *no* downtown.

Having argued that assertion is the *fundamental* pragmatic notion, Brandom goes on to give an account of it in terms of the normative social statuses of *commitment* and *entitlement*. To illustrate these notions, let's work through a hypothetical example of how the game of giving and asking for reasons might be played amongst a group of primitive hominids—or, for that matter, current prairie dogs.[6]

Suppose a creature produces a public token in a social context, and that this act has—if only in that context—the pragmatic significance of explicitly *committing* the creature, in the eyes of its community, to its being the case that *the enemy is approaching*. Each of the other creatures in the group evaluates this commitment, assessing it as correct or incorrect on the basis of their own commitments, explicit or otherwise. In doing so, they take a *normative stance* toward the "speaker", whom the group might then treat as being *entitled* to that commitment. The speaker can be entitled to a claim either by default—e.g., in cases of joint perception or contextually-relevant common knowledge—or by having undertaken *prior* commitments that jointly warrant the one in question. Initially, such normative attitudes are *implicit* in the assessors' overt treatment of the speaker. The group might, for instance, shift its attention in the direction of the speaker's gaze upon hearing 'Enemy!". If the enemy is indeed approaching in a way that is perceptually evident to the group, the entitlement is thereby secured.

[6] I am acutely aware that, despite my efforts at rendering the scenery in a plausibly naturalistic light, the example is not only fictional but transparently artificial in countless respects. I trust these won't matter for the sake of making the key points accessible.

Suppose now that the same speaker, now quite anxious, produces another token—e.g., "Run!" This counts not only as an *explicit* commitment to a (potentially joint) plan of action, but also an *implicit* commitment to the goodness of the inference from "Enemy!" to "Run!" And while the members of the group assessed the speaker as being entitled to the first claim, they may well go on to treat this *new* explicit commitment, i.e., "Run!", as patently unwarranted in the circumstances. Again, in the most basic cases, this "treatment" or "stance" toward the speaker can take the form of overt actions on the part of other group members—e.g., grabbing hold of the speaker and keeping them in place. Another form of response might be the production of overt tokens that commit the group, including the original now-frightened speaker, to an *incompatible* plan—e.g., "Stay!" and "Fight!". This normative incompatibility is itself implicit in the overall communal practices of the group.

From these primitive beginnings, Brandom suggests, a practice can evolve in such a way as to allow for speakers to make *explicit* their commitments regarding the goodness—or, in his terms, "material propriety"—of the inferences that were previously only *implicit* in their practices. Paradigmatically, this is achieved by introducing an expression that has the significance of a conditional. For instance, we can imagine a newly-evolved creature—call it v2.0—that has achieved what Brandom calls "semantic self-consciousness". This involves discursively representing not just enemies and escape strategies, but also *inferential relations between claims*. Creature v2.0 can make explicit its commitment to the goodness of a particular inference by producing a token that has the significance of the conditional, e.g., "If Enemy then Run!" (Note that the force operator, '!', is stripped off from the atomic propositions, "Enemy!" and "Run!", when the latter are embedded in a conditional. This point comes to the foreground in §4.2.) Further elaborations of v2.0's language are manifested with the introduction of new bits of logical vocabulary, all of which serve to express commitments regarding various inferential relations. For instance, negation expresses the relation of inferential *incompatibility* between claims (see below). This is the core thesis of what Brandom calls "logical expressivism".

Turning to more complex logico-semantic devices, consider the phenomenon of indirect discourse. Brandom proposes that *de dicto* reports of "what was said" are used to make explicit the *attribution* of commitments to oneself or to others. For instance, my asserting "Dan said that p" makes explicit *my* commitment to Dan's having explicitly undertaken *his own* commitment to the effect that *p*. In a similar fashion, epistemic vocabulary can play the role of making explicit a speaker's *assessment* of someone's *entitlement* to the commitments that they've undertaken. For instance, were I to upgrade my assertion to "Dan *knows* that *p*," I would not only be *attributing* the commitment to Dan, but also committing *myself* to it, and explicitly representing him as being *entitled* to it. Brandom (2001: ch. 4) points out that these three *pragmatic* aspects of knowledge *attributions*—entitlement-attribution, commitment-undertaking, and commitment-attribution—correspond, in that order, to the three elements of the traditional Justified-True-Belief account of knowledge.

What about deontic modal terms? Brandom argues that these serve to make explicit one's commitment to a plan of action. In saying "I ought to φ", I make explicit my commitment to a plan to φ. Similarly, terms like 'must' can be used to make explicit an inferential propriety that is insensitive to changes in auxiliary assumptions, up to some boundary condition—e.g., natural or legislated laws. Thus, assertions like "In order to light the wick safely, one must first clean one's hands" serve to make explicit a speaker's commitment to the material propriety of the inference from "The wick-lighting is safe" to "Your hands are clean," irrespective of what commitments they have concerning a wide range of possible auxiliary assumptions, such as "It's raining elsewhere," "I never met my grandfather," and "Child-trafficking is a serious problem."

Whether or not these latter are included in one's set of commitments, the inference from "The wick-lighting is safe" to "Your hands are clean" is ostensibly good. Of course, auxiliary commitments that conflict with one's views on *natural laws*—e.g., "Gasoline burns when lit"—would render the inference materially invalid. That's what talk of "boundary conditions" is intended to capture. In the case of the purely nomic reading of 'must', as in "Oxygen must be present for combustion," the inference from "Combustion is occurring" to "Oxygen was present" is good under substitution of virtually any commitment, other than those regarding physical or chemical law.

In this vein, Brandom provides rich pragmatic analyses of a variety of other "vocabularies", including the (meta)semantic devices 'represents' and 'is about', as well as indexicals, alethic modals, anaphoric pronouns, and *de re* attitude reports. Some of the details of these analyses will emerge throughout the present discussion, and I'll devote special attention to his account of *de re* constructions in §3.4. For the moment, the case that's most important to examine is that of 'true', as this bears directly on Brandom's rejection of truth-conditional semantics—arguably the main negative contention that he shares with Pietroski. Brandom offers a clear and well-motivated alternative to the standard treatment of truth in terms of "correspondence"—a notoriously vexed notion that lay at the heart of truth-conditional semantics. Instead he develops a refined version of the "deflationary" approach of truth and reference—arguably the best on the market, as we'll see presently.

§1.3. Deflationism about truth and reference

In asserting that things are thus-and-so, a speaker takes on the pragmatic normative status of a discursive commitment. What is it, then, for *another* person to say that the first speaker's assertion is *true*? The normative pragmatic answer is this: in asserting that some claim is true, one not only *ascribes* a commitment to the speaker who made it, but one also *undertakes that commitment oneself*.[7] This allows for the possibility—enormously useful in social practice—for a speaker to take on commitments that they cannot at present articulate. The inability may be either to memory loss ("I don't remember exactly what she said, but it was definitely true") or to time constraints ("She gave a long speech; I'm not gonna repeat the whole thing now, but everything she said was obviously true"). In the most interesting cases, complete articulation is impossible within physical limits, the set of commitments being literally infinite, as with "The theorems of Peano arithmetic are all true".

Thus, on Brandom's view, the term 'true' and its cognates ('truth', 'correctness', etc.) all serve a distinctive *expressive* function, without which branches of discourse such as mathematics would be impossible. Specifically, these terms all serve to *express a commitment* to something already asserted (or, at any rate, assert*ible*). Brandom thus labels this an "expressivist" account of truth, of a piece with his more general expressivist approach to logical vocabulary. The semantics for truth ascriptions is elaborated still further in light of his discussion of anaphora (§1.7). The notion of inter-sentential (and inter-speaker) anaphora will allow us to appreciate how truth ascriptions can have the pragmatic function of allowing the *inheritance* of commitments and entitlements across inter-personal exchanges.

If this pragmatic expressivist account of our use of 'true' (and related terms) is correct, then there is no obvious reason why anything *further* needs to be said about *truth*. The latter is often

[7] In the special case where the speaker and the assessor are identical—as in, "What I'm saying is *true*!"—the commitment is both redundant and guaranteed, though the term is still useful, in such cases, if only for emphasis and the like.

conceived of as a metaphysical language-world relation—the very one denoted/satisfied by the term 'truth' and its cognates. But there seems to be no explanatory work for which such a relation is *obviously* indispensable—neither in semantics nor, Brandom argues, in any other area of theorizing. This puts his view in the same camp as other versions of "deflationism" about truth—particularly, the well-known disquotational (Quine, 1970), minimalist (Horwich, 1990), and prosentential accounts (Graver, Camp, and Belnap, 1975). I take all of these to share the following core commitments:

(i) a rejection of any account/analysis of truth in terms of correspondence, coherence, or warranted assertibility, on the grounds that truth is *not a relation* (of *any* kind)
(ii) the aim of casting the notion of meaning as *explanatorily prior* to that of truth, both in semantics and elsewhere (including metaphysics, epistemology, and ethics)

The differences between disquotationalism, minimalism, and prosententialism have mostly to do with matters of detail, such as whether to ascribe truth to sentences or to propositions, or how exactly to interpret Tarski biconditionals, liar sentences, and quantified truth ascriptions. These disputes are all strictly irrelevant for our purposes. What's important here is that Brandom's version of deflationism is designed to claim the virtues of each of these prior accounts, without succumbing to the technical objections that have been lodged against them. The three main improvements he suggests are (i) subordinating the semantics of truth ascriptions to his brand of normative pragmatics, (ii) paying closer attention to the *syntax* of truth ascriptions, especially their inter-sentential anaphoric structure (§1.7), and (iii) extending the deflationist account to other semantic notions, including reference, satisfaction, and *de re* representation. While (i) is a straightforward application of Brandom's broader strategy, and (ii) serves largely to immunize his version of deflationism from extant objections, (iii) strikes me as a genuine extension of Brandom's normative pragmatics, allowing it to handle both sentential *and sub*sentential expressions.

The notions of truth and reference are plainly central to the project of truth-conditional semantics. Thus, many have noted that a deflationist account of these notions requires a radical re-thinking of what shape a formal semantic theory should take. In this regard, we now have an embarrassment of riches. In addition to old-school proposals about warranted assertibility, and the pragmatists' short-lived "success semantics" (see Brandom 2009 for critique), we now have the benefit of more modern proposals, including both Pietroski's cognitivist account (§2) and Brandom's inferentialism. Let's examine the latter..

§1.4 Inferentialist Semantics

Having situated assertional practices within the broader sphere of rule-governed social activity, Brandom has introduced his key *pragmatic* notions of commitment and entitlement. He goes on to show how these normative statuses, taken together, can be used to construct a *semantic* theory, whose business it is to explain (in some sense) *how* linguistic expressions can come to play the roles that they do in a community's assertional practices.

In familiar fashion, the explanation goes by way of assigning "meanings" or "semantic values" to expression types. But, in keeping with his other commitments, Brandom does *not* equate meanings with truth conditions, sets of possible worlds, pragmatic success conditions, or assertibility criteria. Rather, he subordinates his semantic theory to the normative pragmatics just outlined, by treating meanings as the *inferential proprieties* that govern the use of linguistic expressions. Slurring over a considerable mass of detail, we can summarize the proposal as follows:

Inferentialism: For a given propositional expression, 'P', the meaning of 'P' can be modeled as the set of sets of other propositional expressions that

(i) entitle one to 'P' in the presence of (various sets of) auxiliary commitments,
(ii) commit one to 'P' in the presence of (various sets of) auxiliary commitments, as well as those to which
(iii) 'P' commits one in the presence of (various sets of) auxiliary commitments, and
(iv) 'P' entitles one in the presence of (various sets of) auxiliary commitments.

A particularly useful *compound* inferential relation turns out to be that of *incompatibility*, wherein taking on one commitment precludes a speaker from becoming entitled to another. In this sense, a commitment to "Herbie is a dog" is incompatible with entitlement to "Herbie is a bird". Brandom (2008, ch. 5) shows how to build a *modal* propositional semantics on the basis of just this incompatibility relation, treating the negation of a claim, for instance, as the minimal set of commitments that are incompatible with it. Here again, the details are illuminating, but only one significant upshot bears highlighting for present purposes.

Casting the meaning of an expression in terms of its inferential proprieties vis-à-vis *other* expressions plainly commits one to meaning holism. A common charge against theories of a holist stripe is that they founder on the rock of compositionality. For instance, Fodor and Lepore (1992) famously argue that inferential roles don't compose, whereas meanings do; *a fortiori*, meanings can't be inferential roles. But the formal incompatibility semantics developed by Brandom (2008) provides a direct counterexample the main premise of this argument, by demonstrating how a inferentialist semantics *can* in fact *provably* meet reasonable compositionality constraints, at least in the modal propositional case.[8] In any event, we will see that there are other reasons to reject Fodor and Lepore's arguments.

§1.5 Substitutional "Syntax"

We've now put on the table both a normative pragmatics and an inferentialist semantics. However, it's relatively uncontroversial that only *proposition*-sized expressions can enter *directly* into inferential relations as premises and conclusions.[9] That being so, we still need to say how *sub*sentential bits of language can have meanings of their own. Identifying subsentential expressions will allow us to explain how such expressions can go on to contribute to the indefinitely many assertions that a creature like us can interpret and produce. While there is no *conceptual* barrier, on Brandom's picture, to a community of creatures/robots using a language with only *finitely* many complex expressions, our own case plainly illustrates that languages can and do come in varieties that admit of *productive* generation. So, while a first-pass presentation of the inferentialist approach is best conducted in terms of a community of creatures that uses a finite language—such as might easily be found in (extra)terrestrial nature or constructed in a robotics laboratory (e.g., AIBO dogs)—it does *not* follow, and is not true, that

[8] Although, as of this writing, an analogous proof for the *quantificational* case remains elusive, I am aware of no principled reasons for thinking that such a proof won't emerge—if not tomorrow, then *someday*. As will become clear throughout, I adopt a resolutely optimistic attitude toward such matters.

[9] For a dissenting view, see Stainton (2006).

the inferentialist program abdicates the responsibility of explaining the productive nature of *some* languages. Quite the contrary; Brandom takes his account of subsentential meanings to constitute one of the core achievements of the inferentialist program.

The primary notion of an inferentialist semantics for subsentential expressions is that of *substitution*, which Brandom inherits from (a reconstructed time-slice of) Frege. Starting with a finite stock of sentence types, Σ, each of whose free-standing (i.e., unembedded) uses have the default pragmatic significance of performing an assertion, we can ask whether any members of Σ can be treated as *substitutional variants* of any others. Keeping to the level of naïve intuition, the sentence 'David admires Herbie' is a substitutional variant of 'Jessica admires Herbie'. We'll see more about how this works in a moment, but the key take-away point is this: if a sentence has a set of substitutional variants, then we can, *to that extent*, discern its subsentential structure. That is, by relating one sentence to another inferentially via substitution, we can notice and distinguish re-combinable subsentential expressions *within* the sentences of the language. Let's work through an example.

Take the sentence 'David admires Herbie' and chop it up any way you like, in respect of phonology, orthography, or whatever surface-level features happen to be relevant to the language at hand.[10] One way of doing so will yield 'Herbie' as a proper part; another yields '… …'erbi'…'. Now do the same with every other member of Σ, where the latter is assumed to be finite.[11] This yields a set of subsentential bits, Γ, consisting largely of nonsense like …'dmire'… and …'vid admi'…. With this in hand, go back to 'David admires Herbie' and substitute any other member of Γ (or, for that matter, Σ) in place of 'Herbie'. You'll find that most such substitutions yield uninterpretable gibberish—i.e., expressions that can enter into no inferential relations with the antecedently interpreted members of Σ. For instance, substituting 'jump' for 'Herbie' yields 'David admires jump', which has no inferential consequences. Same for 'jumps rapidly', 'red', 'we', and …'rential cons'…. By contrast, a commitment to 'Colorless green ideas sleep furiously' would presumably preclude entitlement to 'Nothing ever sleeps furiously', 'There are no colorless green things', 'Ideas can only be red', and many other propositions. There is a clear sense, then, in which this famous sentence is perfectly well interpretable. (It's even false!)

Setting aside gibberish, there will be a subclass of expressions that, when substituted for 'Herbie' in 'David admires Herbie', yield *interpretable* sentences, such as 'David admires Jessica', 'Jessica admires Herbie', 'David feeds Herbie', and 'David feeds Jessica'. (Again, a sentence is interpretable just in case it can play the role of premise or conclusion in an inference.) This subclass of Γ, call it Π, contains all and only the recombinable elements—i.e., the subsentential units of the language—including words, phrases, clauses, morphemes, subjects, predicates, or

[10] We'll do things in terms of orthography here (given the medium), but phonology is plainly the more primitive of the two in the human case, both phylo- and ontogenetically, as textbooks in empirical linguistics have long emphasized. For future robots, the medium will likely be something else—perhaps some descendent of TCP/IP. This would require adapting the substitutional techniques to that particular case.

[11] Any *actual* creature's primary linguistic data (PLD) will, of necessity, be finite for in the course of language acquisition. The obvious analogy to the case of language acquisition in *human* children should not tempt us into assuming that Brandom is pitching an empirical account of the stages of acquisition. Still, the analogy is worth noting, even if we strongly suspect—as generativists do, *pace* Tomasello (2005)—that children's linguistic capacities are productive/generative *right from the get-go*. From the latter hypothesis, it follows that there is no such thing, really, as a finite set of PLD for the child, the child's acquisition device is *always* doing something analogous to hypothesis testing, *even in the absence of input data*. On this picture, the set of PLD is a constantly-moving target—in effect, a massively complex mental representation, or representational structure/system, within the child. The latter is plainly *not* identical with the set of utterances that happened to be produced in a child's presence.

whatever other syntactic categories the language in question contains. We can now call one sentence, S, a *substitutional variant* of another, S*, just in case S is the result of substituting one element of Π with another in any member of Σ. Thus, 'David admires Herbie' is a substitutional variant of 'Jessica admires Herbie', on account of its being the result of the substitution of 'David' for 'Jessica'.

The foregoing puts us in a position to entertain a new inferential relation between sentences. Let's call an inference *substitutional* just in case the conclusion is a substitutional variant of one of the premises. The two inferences, from 'David admires Herbie' to either 'David feeds Herbie' or to 'David admires Jessica', are both fine examples. This notion of a substitutional inference is what allows for an application of the inferentialist strategy to subsentential expressions.

Subsentential Meaning: The meaning of a subsentential expression, α, is the set of *materially good substitution inferences* involving α.

Thus, the meaning of 'Herbie' is the set of inferential proprieties that includes {'David feeds Herbie' ↔ 'David feeds Herb'}, {'David feeds Herbie' ↔ 'David feeds a dog}, and {'David feeds Herbie' ↔ 'David feeds *his* dog'}, and many others[12]. In all such cases, the substitutional inferences are *materially good* in virtue of the fact that 'Herbie' is substituted by any of his other actual names, or by other ways of correctly describing him, uniquely or otherwise.

Needless to say, no one—not I, and certainly not Herbie—will ever have a *full* grasp of the set of inferential proprieties that governs the use of the expression 'Herbie', as this would involve knowing everything there is to know about him. Nor is there any guarantee that any two speakers will converge from the outset on what is correct to infer from "David feeds Herbie"—e.g., whether inferring "David feeds a dog" is (materially) good. Rather, the point is this: given that there are, in point of fact, *plenty* of ways for me to entitle myself to "Herbie is a dog", and no plausible ways (please grant) to undercut that entitlement, it would be *incorrect*, pragmatically *improper*, and epistemically *unwarranted* for someone to assert the opposite. This holds *even if* my interlocutor is *strongly disposed* to maintain a contrary position on the matter (foolishly, no doubt). It's important to always keep in mind that *normative* inferentialism is not about inferential *propensities*; it's about inferential *proprieties*.

§1.6 Predicates and Singular Terms

One consequence of the view presented thus far is that some linguistic expressions can be inferentially stronger or weaker than others. Consider the verbs 'runs' and 'moves'. The latter is logically stronger than the former because all substitution inferences from 'x runs' to 'x moves' are good, but the reverse inferences generally aren't. In such cases, the substitution inferences are said to be *asymmetric*. We also find terms that *invariably* enter into *symmetric* substitution inferences—e.g., from 'Mark Twain was an American' to 'Samuel Clemens was an American' and back again. To make the latter type of inference explicit, subsentential expressions of identity and nonidentity can be introduced, yielding propositions of the form $\alpha=\beta$ and $\alpha\neq\beta$ (e.g., 'Sam Clemens is identical with Mark Twain' and 'David is not Herbie').[13]

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[12] For simplicity of presentation, I suppress issues to do with possessives like 'my' and 'his', and indexical expressions more generally. Brandom (1994, 2008) supplies an account of these, but the details are irrelevant here.

[13] The notion of "introduction" that I intend here is the one developed in Brandom (2008). Roughly, a community is capable of introducing a novel expression, in this sense, just in case its members already have the *practical* abilities that are necessary and sufficient for being able to express—i.e., to make explicit—normative attitudes that

As we will see in §4, Brandom holds that the distinction between predicates and singular terms comes down to the distinction between those expressions that *must* license *only* symmetric inferences (e.g., 'Herbie' and 'the dog'), and those that merely *can* license symmetric inferences, but *need not* do so (e.g., 'deer-like', 'jumps', and 'rapidly). On the basis of this claim, Brandom goes on to develop a complex line of reasoning whose ultimate conclusion I'll call the "asymmetry constraint".

Asymmetry Constraint: Any language that draws no distinction between predicates and singular terms (conceived in the above manner) *is in principle precluded from introducing conditionals*—i.e., expressions that make explicit one's commitment to the goodness of an inference—and other basic operators of propositional logic.

This claim will come to the foreground when we contrast it with Pietroski's predicativism, according to which there are in fact no singular terms at all in natural languages. If Brandom's argument succeeds, then Pietroski's predicativist semantic theory faces a serious challenge. Contrapositively, if Pietroski's predicativism is correct, then there must be a flaw in Brandom's reasoning. This is, in fact, the final puzzle—not to say *mystery*!—of the overall reconciliation project that I'll be urging here.

§1.7 Types, Tokens, and Anaphoric Chains

The expressions discussed thus far have all been linguistic *types*, tokens of which may well diverge in meaning from their primary significance in the language. Indeed, terms like 'Herbie' have so many different uses—one for my dog, another for the pianist, Herbie Hancock, and countless others—that Brandom needs an account of what makes any use of 'Herbie' *semantically co-typical* with any other. The question applies even to *intra*-sentential occurrences: What makes it the case, for instance, that both tokens of 'Herbie' in 'Herbie admires Herbie' of the same type in a given communicative context?

In providing his answer, Brandom introduces the last of the major technical notions that he needs in order to carry off his overall project—viz., the notion of *anaphora*. Linguists and philosophers have paid a great deal of attention to *intra*-sentential anaphora, as in 'If a man is a police officer, then he was born out of wedlock', where the pronoun 'he' is anaphoric on 'a man'. Syntacticians, in particular, have devised principles of generative grammar that aim to explain the natural distribution of anaphoric expressions within sentences of natural language. Somewhat less effort has thus far been expended on analyzing *inter*-sentential anaphora, as in the following exchange between speakers Mihir and Rushal.

> Mihir: That man seems to have fallen ill right after he approached the police line.
> Rushal: He must have gotten hit by their fancy new sonic weapon.
> Mihir: Oh, hey, I didn't see you there! Do you happen to know the guy?
> Rushal: No, I just heard you talking about him and I figured I'd chime in.

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were previously only implicit in their practice. Thus, the practical ability to implicitly treat someone as having entitled themselves to q by committing themselves to p is both necessary and sufficient to introduce conditional expressions that make explicit the material goodness of that inference—e.g., '$p \supset q$' and 'If p then q'. We will see in §2.5 that Pietroski's notion of concept introduction is different from Brandom's, and arguably orthogonal.

Here, an anaphoric chain is initiated by Mihir's use of 'That man', which is then picked up by 'he' later in the same sentence. But the chain doesn't end there. Rushal's use of 'He' is anaphoric on Mihir's use of 'That man' and 'he'. Mihir's response picks up the anaphoric chain with an occurrence of 'the guy', which then continues onward to Rushal's use of 'him', and to occurrences of other expressions in subsequent discourse. Setting aside syntactic issues, what can we say about this phenomenon at the level of *meaning*?

In keeping with his inferentialist semantics, Brandom argues that an anaphoric chain is one in which the inferential proprieties governing the anaphoric initiator (e.g., Mihir's use of 'That man') are *inherited* by subsequent expressions in the chain. Thus, if Mihir's use of 'That man' is partly governed by his commitment to 'That man is falling ill on live television', then Rushal inherits this commitment (among others) in picking up the anaphoric chain with the use of 'He', along with whatever entitlements for this claim Mihir had already secured prior to Rushal's appearance on the scene.

With this account in hand, Brandom treats as a special case occurrences that are treated as semantically co-typical *because* they are phonologically or orthographically co-typical—e.g., the two occurrences of 'Herbie' in 'Herbie admires Herbie'. From this perspective, *all* expression types consist of long-stretching anaphoric chains of individual use—an idea familiar from causal theories of reference-borrowing, though shorn of various optional commitments. This account also makes it clear what's happening at the level of *pragmatics*. In picking up anaphoric chains, speakers are able to take on normative statuses—paradigmatically, commitments and entitlements—without themselves having *explicitly avowed* those statuses, and often without having much (if any) idea what exactly it is that they've inherited. To illustrate, we can extend the above example.

Suppose Rushal had no prior commitments regarding the victim's appearance on television, or indeed anything at all about the victim, but was strongly committed to the claim that police don't use sonic weapons on camera. In that case, upon being subsequently apprised of Mihir's entitlement to 'That man is falling ill on live television', Rushal will be under normative pressure to either revise his prior commitments about on-camera police violence, or to withdraw the claim that the victim *must* have been affected by a specifically *sonic* weapon. In this second case, the revision can target either the predicate 'sonic'—perhaps the police used an invisible gas—or the alethic modal expression ,'must'. The latter, on Brandom's view, functions to make explicit the *modal robustness* of an inference—i.e., its insensitivity to substitutions of background auxiliary commitments, up to some boundary conditions (e.g., physical law). In the present case, the boundary conditions are set by Rushal's commitments regarding the general institutional practices of local police. In order to regain epistemic equilibrium, Rushal can revise various commitments concerning these practices; for instance, he might conclude that the local Sheriff has deemed this to be a special occasion, on which on-camera use of sonic weapons is warranted.

§1.8 Summary

We've now surveyed the main contours of Brandom's overall philosophical project. The explanatory strategy he pursues can be characterized as "top-down", in the sense that he begins by offering an account of communal normative practices, in the broadest sense, and identifies within these an important subclass—namely, practices that serve to institute distinctively *linguistic* norms governing assertion and other communicative acts. (One last plea for demands!) Such norms pertain to the inferential proprieties that expression types have in their semantically primary occurrences. Thus, the account moves "down" a step—from a normative pragmatics that posits statuses of commitment and entitlement, to an inferentialist semantics that aims to analyze *meaning* in *terms of* these statuses. The meaning of a propositional expression type is, on this picture, identified with its normative inferential role—i.e., what *other* claims it commits or entitles one to, and what commitments one must undertake in order secure an entitlement to it.

Drilling down still further, Brandom develops the substitutional approach, which allows one to "dissect" proposition-sized expression types, revealing subsentential bits of vocabulary. These carry their own "ingredient content", despite lacking the *free-standing* significance of propositional expressions that enter directly into inferences as premises or conclusions. The details of this proposal put in place the theoretical commitments that Brandom needs in order to distinguish predicates from singular terms—a distinction that he goes on to argue will be discernable in *any* linguistic practice that allows for the introduction of conditionals and other logical operators (§4.3).

Having offered a treatment of propositional and subsentential expression types, Brandom steps down another rung on the explanatory ladder, developing a conception of *anaphora* that applies far more broadly than standard discussions in the literature might lead one to suspect. The anaphoric relationship is, on this view, one of *inferential inheritance*, wherein the proprieties governing the use of one expression—the initiator of an anaphoric chain—are taken to then *also* govern the expressions occurring later in the chain, irrespective of the speaker's acknowledgement (or even awareness) of the statuses they've thereby undertaken. The latter condition serves to explain how speakers can felicitously use expressions whose total set of inferential proprieties is *unknown* to them, and perhaps even to *anyone* in the community.

One might think that all of this is utterly wrongheaded right from the get-go—the normativity, the substitutions, and even the top-level goal of delineating language-use *as such*. Indeed, from the perspective of a mainstream contemporary linguist or philosopher of language, Brandom's whole "top-down" explanatory strategy will seem downright perverse. The more common *bottom-up* alternative goes as follows.

Taking for granted the notions of denotation/reference and satisfaction, as applied to subsentential expressions, the bottom-up theorist seeks to formalize a compositional apparatus for building propositions out of them. Free-standing propositional complexes are thereby recursively assigned their own special kind of semantic value: e.g., possible-worlds truth conditions (Heim and Kratzer, 1998) or sets of possible worlds (Stalnaker, 1984). This, in turn, opens the door to a theory of linguistic *communication*, according to which speakers append illocutionary forces to the range of recursively-specified meanings, yielding a variety of speech-act types (questions, commands, etc.). The inferences in which a (now-interpreted) speech act type figures can then be classified as good or bad in virtue of the semantic structures that the combinatorial apparatus assigns to their premises and conclusions, as well as the illocutionary forces that (somehow) "attach" to those structures.

Having thus analyzed the semantic properties of speech acts and inferences, one might note that some—perhaps, in the end, *all*—of these have features that reliably trigger unencapsulated pragmatic reasoning. This motivates the familiar project of supplementing a pragmatic theory with "maxims" of rational cooperative communication/action (Grice 1989; Sperber and Wilson, 1986). Theorists who have carried out this latter project have developed impressive accounts of implicature, metaphor, and other complex communicative phenomena (Levinson, 1983; Harris, 2020).

Proponents of the bottom-up strategy have pressed a catalogue of objections to Brandom's project. These include, but are not limited to, the following: (i) insistence on a *compositionality* constraint that the inferentialist allegedly can't accommodate; (ii) rejection of the idea that language is fundamentally a *communicative* system; (iii) requirement that any *legitimate* inquiry foreswear trafficking in normative assessments; and (iv) an allegation to the effect that normative inferentialism is incompatible with what is known empirically about the human mind/brain, particularly in respect of its language-processing abilities.

Before any of these challenges can be met, each stands in need of careful articulation. As previously noted, I believe that such a task is best undertaken by pitting Brandom's project against what appears, at first blush, to be a rival alternative. (As advertised, I'll argue afterwards that the appearances are often deceiving in this regard.) With that in mind, I now turn to the work of Paul Pietroski, whose semantic theory is a recent and powerful contribution to the larger enterprise of generative linguistics.

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§2. Pietroski: Meanings as Pronounceable Instructions for Concept Assembly

The theoretical commitments that comprise Paul Pietroski's approach to natural-language semantics are advanced and defended in his recent book, *Conjoining Meanings* (henceforth *CM*).[14] In this section, I summarize several of Pietroski's main contributions, highlighting aspects of his view that bear on my ecumenical strategy in §§3-4. To be clear from the outset, the ideas laid out in *CM* strike me as constituting genuine progress in our understanding of the psychological mechanisms of human language use. Moreover, I find wholly compelling his arguments against the central pillars of received views in semantics—particularly, the commitment to an extensional/truth-conditional approach. The book, overall, is replete with rich and instructive discussions of topics that go well beyond the scope of the present discussion. But while we won't be able to look at the details of some of Pietroski's original proposals here, it's worth noting that they are all, to my mind, persuasively motivated by historical, formal, and empirical considerations. That having been said, let's dive in.

§2.1 A different methodology and new explanatory aims

While Brandom's inferentialist approach is virtually unknown in cognitive science, the methodology of generative linguistics will be familiar to many in the field, at least in broad outline. Rooted in a foundational commitment to *naturalistic* inquiry, the idea is to treat language as a biological phenomenon—not necessarily in the sense that it has an adaptive function (Chomsky [2016] disputes this), but in the sense that a neurophysiologically realized cognitive structure is the explicit target of inquiry. The linguist thus works on the assumption that human minds contain a language-specific device—a "faculty", "module", or "mental organ"—with a distinctive computational architecture, a proprietary representational format, and dedicated/domain-specific information-processing routines. The goal is to provide a detailed specification of each of these, yielding a neurocognitive account of the acquisition and use of language.

[14] This section elaborates the material in Pereplyotchik (2019). The operative notion of a *subpersonal* level of description is spelled out in Pereplyotchik (2017: ch. 7).

On analogy with bodily organs, the faculty of language (henceforth FL) is assumed to "grow" within the child during the early years of development. This happens in accordance with a genetic program, phenotypically realized in the child's innate ability to acquire linguistic competence under a diverse range of social and environmental circumstances. Thus, a central aim of generative linguistics is to specify not only the grammar of an adult language, but also the principles that underlie language *acquisition*—particularly those that allow the child to home in on a *specific* grammar in a relatively short time, with little or no (overt) negative evidence (Chomsky, 1986; Yang, 2006). This problem is made exceedingly challenging by the fact that natural languages are invariably productive/generative, meaning that they allow for boundless applications of combinatorial recursive operations, yielding a discrete infinity of nonredundant[15] interpretable structures.

The generativist's strategy for dealing with this central feature of natural language is to posit grammatical principles that are inherently *compositional* at all levels of analysis—phonology, morphology, syntax, and semantics. The syntactic module of FL is taken to merge the elements of the lexicon—atomic units of a language that contribute their distinctive meanings to more complex structures. On the basis of these, the semantic module recursively generates complex *meanings*, which can enter into downstream personal-level cognition—judgment, reasoning, planning, and the like.[16]

Pietroski's main goal in CM is to characterize the semantic module by offering a detailed proposal about its proprietary representational format—specifically, the nature of the lexical items—and the computational operations that assemble larger interpretable structures. At the level of format, the hypothesis he develops is that virtually all lexical items are predicates, the latter being restricted to only two types—monadic and (semi)dyadic. Regarding computational operations, Pietroski aims to make do with a bare minimum of compositional semantic principles, with the lion's share of work being done by nothing more than two flavors of predicate conjunction (one for each type of predicate). We'll look at some of the details shortly, maintaining our present focus on matters of methodology.

Following Chomsky (1986, 1995, 2000), Pietroski adopts an *individualist* position, taking the object of study to be an "I-language"—an *i*ntensionally-specified procedure *i*nternal to an *i*ndividual language user. He supports this with forceful arguments against the alternative conception of language(s) that we find in the work of David Lewis (1969, 1970, 1973). A language, on this rival picture, is a kind of abstract object—namely, an extensionally-specified a set of well-formed sentences—which is "selected" by a population of creatures, via the adoption of social/communicative "conventions". The latter Lewis sees as jointly constituting a *public* language, such as English or Norwegian—what generativists refer to as "E-languages". Pietroski rejects virtually every aspect of this picture. We'll look at his reasons for doing so in §4. For now, it's sufficient to distinguish three key points of contention.

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[15] Pietroski points out that this goes well beyond mere recursion, which is trivially satisfied by any languages with a rule for applying sentential operators. The infinitude of English thus differs *qualitatively* from the infinitude of a language that permits the formation only of P, P&P, P&(P&P),… or P, ~P, ~~P, ~~~P,….

[16] It's important to note that what has been said thus far is not (yet) intended as a theory of real-time/on-line language processing. Rather, it is to be seen as an abstract characterization of the architecture and internal operations of a specific cognitive structure, acquired at birth and persisting in a stable state thereafter (Chomsky, 1995).

First, there's the metaphysics. Lewis (1973) says languages are abstracta, whereas Pietroski sees them as biologically-instantiated computational procedures. Then there's the issue of extensionality. Pietroski rejects Lewis's theoretical goals, which consist merely of extensionally specifying meaning-pronunciation pairs, and adopts instead a more weighty explanatory aim—namely, that of specifying human linguistic competence as a function-in-*intension*. Only in this way, he argues, can the resulting theory capture the *psychologically real* operations that yield interpretable structures. Finally, there's the issue of publicity, and related troubles with Lewis's notion of "selection". Pietroski's individualist stance leads him to eschew the folk-ontological commitment to public languages, at least for the purposes of mature empirical inquiry. This manifests in his methodological practice of focusing on matters of individual psychology—e.g., internal mechanisms of semantic composition—rather than the social practices of linguistic communication. Accordingly, Pietroski sees Lewis's appeal to public conventions as generally unhelpful for—indeed, an outright *distraction from*—the empirical study of linguistic meaning.

Pietroski's disagreements with Lewis go well beyond such methodological issues, extending to matters of technical detail. For, in addition to the large-scale commitments mentioned thus far, Lewis (1970) also developed a powerful formal apparatus for conducting semantic theorizing. Expressions, in this scheme, are assigned "semantic types", which are either basic or recursively derived. The interpretation of complex structures is then accomplished by functions that map one semantic type onto another. In its most familiar version, such a semantic theory will assign sentences the basic type <t> and singular terms the basic type <e>. Thereafter, monadic predicates can be treated as having the derived type <<e>, <t>>, which is a function from things of type <e> to things of type <t>.

Although this formal typology presupposes no particular metaphysics or metasemantics, it's common in practice to think of singular terms as denoting entities (e.g., Jessica), and sentences as denoting truth-values (T and F). With this in place, monadic predicates like 'swims' can be assigned the semantic function of mapping the entities in its domain to the truth-values in its range. For instance, 'Jessica swims' is mapped to T just in case Jessica (the actual person) satisfies the predicate 'swims'; otherwise, F. Likewise, adverbs such as 'often' and 'expertly' have the derived type <<<e>, <t>>, <t>>, which is a function that maps the semantic value of predicates (i.e., functions from <e> to <t>) to the semantic values of sentences (i.e., T or F). Put somewhat imprecisely, the intuitive idea is that 'Jessica swims expertly' is mapped to T just in case 'expertly' is satisfied by the predicate 'swims' when applied to 'Jessica'.

It's no exaggeration to say that this general framework is seen as a foundational contribution to formal semantics, even by generative linguists who have no truck with—or, indeed, no awareness of—Lewis's broader projects. Part of what makes Pietroski's negative contentions so radical, then, is that he rejects wholesale this now-mainstream approach to semantic theorizing. In particular, he argues that taking an infinite hierarchy of types as explanatorily primitive is not only unparsimonious, but leaves wholly unexplained crucial aspects of the natural languages that children invariably acquire. As a matter of empirical fact, humans language permits the construction of only a *limited* class of semantic types, not the infinite range of logically possible ones. This empirical generalization plainly stands in need of explanation, which a semantic theory can't provide if it takes all possible types as available to a speaker right from the start.

> One can say that thinkers must have the requisite abstractive powers, given the capacities required to form thoughts like ABOVE(FIDO, VENUS) & BETWEEN (SADIE, BESSIE, VENUS). But one needs an account of these alleged powers—which permit abstraction of a tetradic concept from ABOVE(_, _) and BETWEEN(_, _, _)—to explain how thinkers can form the concepts that *Begriffsschift* expressions reflect. This is not to doubt the utility of Frege's

> logical syntax. On the contrary, his proposals about the architecture of thoughts were major contributions. But Frege insightfully invented a logical syntax whose intended interpretation raised important questions that he did not answer.
>
> One can insist that given any polyadic concept with n unsaturated "slots," a human thinker can use n-1 saturaters to create a monadic concept, leaving any one of the slots unfilled. But that leaves the question of how we came to have this impressive capacity. And in chapter six, I offer evidence that a simple form of conjunction lies at the core of unbounded cognitive productivity. Our natural capacities to combine concepts are impressive, but constrained in ways that suggest less than an ideal Fregean mind.

Pietroski recommends a more parsimonious alternative—one that eschews the infinite hierarchy of semantic types and posits only a very small handful, including, most importantly, monadic and quasi-dyadic predicates. "The idea [is] that with help from Slang syntax, we can generate an analog of GIVE(VENUS, _, BESSIE) without saturating GIVE(_, _, _)—much less saturating it twice, or thrice, and then desaturating once" (103).

Nor does his iconoclasm end there. As noted earlier, Lewis's general framework for semantic theorizing leaves open a variety of issues in metaphysics and metasemantics. An equally mainstream approach to natural-language semantics is decidedly more committal on these points. Donald Davidson's truth-theoretic semantics (Davidson, 1983), as well as the many variants of it that have now been developed, identifies the meanings of linguistic expressions with their *extensions*. Thus, truth conditions (perhaps relativized to possible worlds) are seen as the semantic values of sentences; entities are the values of singular terms; sets are the values of predicates; events in the case of verbs, and so on. Pietroski marshals a battery of arguments against this familiar approach. We'll examine these shortly. For now, we note only that this anti-extensionalism is a core commitment that he shares with Brandom. It is, therefore, a major plank in the bridge that I aim to build between the two in §§3-4.

§2.3 Meanings are definitely not *extensions*

Pietroski sees semantics as a naturalistic inquiry into "how Slang expressions are related to human concepts" (115). Some theorists wish to simply *identify* meanings with concepts, but Pietroski points out that this leaves wholly unexplained the psychological processes that *constitute* our semantic competence. I'll argue in §4 that this point applies to Brandom, who sometimes speaks indiscriminately of meanings, concepts, conceptual contents, intentional contents, discursive contents, propositional contents, and so on. However, as I'll emphasize there, the difference can only be viewed as a substantive theoretical *dispute* if we let their use of the folk term 'meaning' bewitch us into assuming that they have a common explanatory target, contrary to fact.

Better, I think, to appreciate the highly theoretical nature of this piece of jargon and the different—*but not thereby incompatible*—explanatory goals of the two frameworks in which it shows up. Thus, we can distinguish meaning_B from meaning_P and proceed to contemplate how the two are related, this now being a jointly philosophical and empirical question, not a boring verbal one. Indeed, this point is made explicitly by both Pietroski and Brandom, in connection with both 'meaning' and another vexed notion—that of 'concepts'—which notoriously plays a *wide* variety of roles in diverse research contexts. Here again, we can speak of concepts_B and concepts_P, aiming to articulate the relations between them. Likewise for 'thought', 'judgment', and other terms, when explicit disambiguation is required. (See also the discussion of the notorious '-ing/-ed' ambiguity in §4.2.)

As noted above, another popular idea is to identify meanings with extensions (Davidson, 1983). The central negative contention of *CM* is that the notions of extension, truth, and denotation should play *no explanatory role* in a psychologically-oriented semantics for natural languages ("Slangs"). Pietroski argues persuasively that the best empirical theory of the relation between Slang expressions and concepts will *not* identify meanings with extensions. Indeed, he rejects even the weaker claim that meanings *determine* extensions. He proposes, instead, to identify meanings with something entirely different—in particular, something that can play the psychological role of relating language to cognition. The candidate he recommends is this: *pronounceable instructions for accessing and assembling concepts*. We'll look at this in some detail, but let's first get clear on *why* Pietroski rejects the truth-conditional orthodoxy that dominates formal semantics. As we'll see, there are a great many reasons. To my mind, no one of these is necessarily decisive, but, taken together, they strongly suggest turning away from the extensionalist project and starting anew, *however* much revision this might require. As we go along, I'll land a few jabs of my own, but Pietroski's are by far the weightier blows.

§2.3 Objections to truth-conditional semantics

Pietroski views truth-conditional semantics (henceforth 'TCS') as an empirical hypothesis about Slang expressions, according to which there is a relation—call it "true of", "refers to", "denotes", or whatever you like—that holds between words and items in the world. TCS views this relation as being of central importance to our theoretical characterization of natural-language meanings. In rejecting this hypothesis, one need not deny, of course, that there are words or that there objects (e.g., babies and 'bathwater'). One can, instead, deny that there is a unique relation between them, let alone one that's suited to playing the theoretical role of *linguistic meaning*. Here is how Pietroski puts the point:

> I don't think 'sky' is true of skies (or of sky), much less blue skies or night skies. I don't deny that there are chases, and that in this sense, chases exist even if skies don't. But the existence of chases doesn't show that 'chase' is true of them... [Likewise], there is no entity that 'Venice' denotes. In this respect, 'Venice' is like 'Vulcan', even though one can visit Venice but not Vulcan... I also agree that there is a sense in which there are blue skies, but no blue unicorns. But it doesn't follow that 'sky' is true of some things, at least not in the sense of 'true' that matters for a theory of truth... [T]here is no call to quantify over skies, in physics or linguistics. (68)

As the example of 'Vulcan' illustrates, words can perfectly well be meaningful without having extensions. Pietroski's view is that this holds of *all* Slang expressions. What's interesting about words like 'Vulcan' is they "illustrate the general point that words don't *have* extensions". The idea isn't merely such terms have *empty* extensions; it's that they have *none at all*.

Even if words *did* have extensions, the latter couldn't be identified with meanings, if only because "expressions with different meanings can have the same 'extension'" (15). Fans of TCS will typically appeal to "non-actual possibilities" in dealing with this issue. For instance, 'unicorn' and 'ghost' are said to have the same extension in the *actual* world, but they differ in meaning—the reply goes—because they have *different* extensions in *other* possible worlds. Pietroski correctly points out that this "is an odd way to maintain that meanings are extensions."

> If the meaning of a word is not whatever set of things that the word happens to be true of, why think the meaning is a mapping from each possible world w to whatever set of things that the word happens to be true of at w? [If] Slang expressions need not connect pronunciations to actual things, it seems contrived to insist that these expressions connect pronunciations to possible things... [I]nvoking possible unicorns is contrivance on stilts. (12)

Doubtless, fans of TCS will see this as little more than an *ad hominem*. We'll look at stronger arguments shortly. For now, I want to emphasize that this point—or, in any case, a version of it—carries more weight than is commonly appreciated. Let me take a brief aside to develop it in my own terms.

The intuitive considerations that motivate TCS (e.g., for introductory semantics students) almost always have to do with objects that are available for perceptual inspection. ('David' refers to *this guy*, 'my desk' refers to *that thing*, and so on.) This serves to illustrate, *at the level of pre-theoretical intuition*, how linguistic expressions "hook onto the world"—namely by way of *perceptual contact* (indeed, *literal* contact, in the case of haptic perception). Shortly thereafter, the details of one or another formal theory are introduced, giving the student little time to reflect on how far the initial illustration can plausibly generalize. (Spoiler alert: not very far!) If philosophical questions happen to arise about the status of these "reference" and "correspondence" "relations"—e.g., with regard to empty names and predicates ('Vulcan', 'unicorn', etc.)—the instructor can use the opportunity to explore various technical proposals for dealing with such "special cases"—e.g., Russell's theory of names as disguised descriptions, or the formal apparatus of possible-worlds semantics. Attention is thus deflected away from how massive the intuitive problem really is. Here's a much-needed corrective.

Consider for a moment the vast range of expressions that we can readily produce and comprehend, and reflect on how vanishingly few of these have anything much to do with what's going on in physical reality, let alone with things that we can perceptually inspect in any intuitive sense. We speak of Santa and his elves, gods and demons, goals and fears, opportunities and temptations, aliens and chem-trails, reptiles and unicorns, futures and fictions, numbers and functions, nouns and verbs, fonts and meanings, haircuts and field-goals, stocks and derivatives, mergers and monopolies, economies and governments, boson fields and spin-foams, black holes and electrons, Blacks and whites, Jews and Frenchmen, London and Moscow, classes and genders, protests and stereotypes, jocks and nerds, bits and bytes, poems and operas, humor and beauty, and even the possibility (albeit dim) of true liberatory justice.

Appreciating the sheer scope of the phenomenon to be explained renders, to my mind, *utterly implausible* the strategy of taking direct perceptual contact with the world as our model of how language relates to reality *in general*. Moreover, the total lack of convergence that we find amongst metasemanticists when we go looking for a metaphysical account of truth and reference—conceived of, again, as a Very Special sort of natural relation—strikes me as further grounds for abandoning the project of extensional semantics immediately and forthwith. It helps, of course, that Pietroski supplies a powerful *alternative* framework for doing semantics. And it certainly doesn't hurt, I reckon, that Brandom complements this with an independently attractive ("deflationist") account of truth and reference.

All that aside, Pietroski has a further, more powerful argument against invoking non-actual possibilities for the purpose of individuating meanings. He makes use of Kripke's contention that the non-existence of unicorns in the actual world implies their non-existence in *all other* possible worlds (Kripke, 1980). Of course, there may well be creatures in other possible worlds that *look* a lot like what we imagine unicorns would look like. But they would not thereby *be* unicorns, and our word 'unicorn' would not thereby be true of them. If that's correct, then 'ghost' and 'unicorn' aren't just co-extensive in *our* world; they're co-extensive in *every* possible world. Thus, no identification of meanings with extensions, actual or possible, will distinguish the meanings of those two words. Likewise for all of the related cases—empty names, defective predicates, necessary falsehoods, and so on.

One might reply by rejecting Kripke's semantic and metaphysical assumptions, and adopting instead a Lewisian counterpart theory, but Pietroski points out several problems for this strategy as well. Adopting the terms 'LUNICORN' for Lewisian unicorn-lookalikes and 'KUNICORN' for the whatever it is that Kripke has in mind, he makes the following powerful retort.

> We can grant that some theorists sometimes use 'unicorn' to express the technical concept LUNICORN. But if 'unicorn' can also be used to express the concept KUNICORN, then it seems like contrivance to insist that the Slang expression has a meaning that maps some contexts onto the extension of LUNICORN and other contexts onto the extension of KUNICORN. If we assume that words like 'possibly' have extensions, then perhaps we should specify the meanings of such words in terms of a suitably generic notion of world that allows for special cases corresponding to metaphysical and epistemic modalities; cp. Kratzer. But in my view, theorists should not posit (things that include) unicorns in order to accommodate correct uses of 'Possibly/Perhaps/Maybe unicorns exist' or 'There may be unicorns'; and likewise for squarable circles.

Thereafter, the dialectic turns to matters that we need not enter into here. Suffice it to say that, even if this worry about fine-grained meanings can ultimately be defused, TCS would still face Pietroski's more technical (and potentially more damaging) objections. These include matters pertaining to liar-sentences, as well as the more widespread and natural phenomenon of event descriptions. Sadly, these too go beyond the scope of our discussion. One argument that I do want to say a bit more about, though, is on the topic of, where Pietroski's view of the matter finds wide acceptance among generative linguists—though, notably, not philosophers of language (see, e.g., Michael Devitt's paper in this issue.)

Following Chomsky (2000), Pietroski points out that 'water' is polysemously used to talk about many substances—those found in wells, rivers, taps, etc.—nearly all of which have *lower* H_2O contents than substances that, at least *prima facie*, are *not* water, including coffee, tea, and cola (*CM*, 21). This presents a challenge to theories that view 'water' as bearing a reference relation to (all instances of?) the natural kind *water*, whose metaphysically essential property is *being composed of H_2O molecules* (Kripke, 1980). If coffee, tea, and cola all have more H_2O in them than most ordinary instances of water, then it's not clear why 'water' doesn't bear the reference relation to *them*, rather to the stuff in the local rivers and wells.

A related consideration has to do with predicate conjunction. The word 'France' can be used in expressing either of two concepts: FRANCE:BORDER and FRANCE:POLIS. The border is hexagonal and the polis is a republic. But, Pietroski points out, the polysemy of 'France' "does not imply that something is both hexagonal *and* a republic, much less that 'France' denotes such a thing" (74). Similarly, while 'London' can be used to talk about "a particular location or a polis that could be relocated elsewhere," it is plain that "no location can be moved, and no political institution is a location." Pietroski concludes that "no entity is the denotation (or 'semantic value') of 'London'; the ordinary word *has* no denotation" (73, emphasis mine).

§2.4 Meanings as pronounceable instructions

Let's turn now to Pietroski's positive views. As noted earlier, the main goal of *CM* is to defend the hypothesis that linguistic meanings are "pronounceable instructions for how to access and assemble concepts" (1). More specifically,

> each lexical meaning is an instruction for how to access a concept from a certain address, which may be shared by a family of concepts. … A Slang expression Σ can be used to access/build/express a concept C that is less flexible than Σ—in terms of what Σ can be used to talk about, and how it can combine with other expressions, compared with what C can be used to think about and how it can combine with other concepts— since Σ might be used to access/build/express a related but distinct concept C* .

Unpacking Pietroski's hypothesis requires getting clear on the three key notions of *pronounceable instructions*, *compositional assembly*, and *conceptual types*. Each is more challenging than the last, so we'll start with instructions and work our way up.

§2.4.1 Pronounceable instructions

An utterance of a sentence is a spatiotemporally located event, in which a speaker produces a physical signal. The latter serves, on Pietroski's view, as an instruction for the hearer's FL to perform a computational procedure.[17] The instruction can be carried out by any hearer whose I-language is sufficiently similar to the speaker's. The acoustic properties of an utterance, upon being transduced, trigger an early perceptual constancy effect, whereby a dedicated module imposes phonological categories on the neural encoding of the acoustic blast. These cognitive operations serve, in turn, as instructions for the further segmentation of the phonological units into syllables and eventually into morphemes and other lexical items. The latter, on Pietroski's view are best seen as instructions for accessing ("fetching") individual concepts, which he conceives of as atomic units of one or another language of thought. I say "one or another" because his view leaves open the possibility, which he goes on to explore and even endorse, that there are *many* languages in which the mind conducts its information-processing. We'll return to this point in connection with Pietroski's discussions of Frege (§2.5).

Importantly, Pietroski maintains that concepts reside in semantic "families", which have their own "addresses" in a broader cognitive architecture. This is a large part of his explanation of the aforementioned phenomenon of polysemy. The idea is that one and the same lexical item can be an instruction for fetching "a concept from a certain lexical address … shared by a *family* of concepts" (8). Because a lexical instruction points only at an address, rather than a specific concept, it's left open for *downstream* processing routines to determine which particular concept from the indicated address/family is "relevant" in the present context.

This, of course, raises deep and difficult questions about how hearers manage this latter step—i.e., reliably accessing the relevant concept(s) in a given context, rather than the irrelevant ones from the same conceptual family. What psychological mechanisms select just *one* of a family of concepts residing at a common lexical address? In large part, Pietroski leaves this issue open—justifiably so, given everything else he's juggling. But it's worth remarking in the present context that the mechanisms of this kind of selection are widely agreed to involve—indeed, to *require*—precisely the kind of nondemonstrative pragmatic reasoning that Brandom has argued to be constitutive of conceptual contents.

[17] "I hope the analogy to elementary computer programs, which can be compiled and implemented, makes the operative notion of instruction tolerably clear and unobjectionable in the present context. … Instructions can take many forms, including strings of '0's and '1's that get used—as in a von Neumann machine—to access other such strings and perform certain operations on them. … And instead of arithmetic operations that are performed on numbers, one can imagine combinatorial operations that are performed on concepts" (108).

§2.4.2 Assembling concepts

Turn now to the second key notion in Pietroski's main hypothesis—viz., the compositional assembly of concepts. In general, instructions for assembling something can vary along any number of dimensions. Some are clear; some aren't. Some are detailed; others are vague. Some are simple; others are complex—i.e., composed of simpler instructions. Moreover, not everything to which an instruction is presented is capable of carrying it out. Some computers can't run the software that others can. Some chefs can't bake the cakes that others have no trouble baking. And some proteins (or cells) can follow genetic instructions that others simply can't. Lastly, the *products* of successfully carrying out instructions can vary widely. The same student, with the same instructions, can succeed or fail on an exam, depending on whether they've had sleep the night before. Likewise, a novice barista will generally make worse coffee with low-quality ingredients than with high-quality ones, successfully following the same instructions both times.

Given that the semantic module of FL is assumed to have a stable processing routines, carried out in a proprietary representational format, it follows that it won't be able to process just *any* old instruction, but only a restricted kind. Likewise, it will only be capable of assembling only a limited class of outputs. The question, then, is what kinds of instructions the semantic module is capable of implementing and what sort of structures it's capable of building.

Many theorists aim at capturing something called "compositionality"—a piece of theoretical jargon that, perhaps more than most, has been worn smooth by a thousand tongues (to use Wilfrid Sellars's clever phrase). Of the many ways of cashing it out, Pietroski maintains that what's required for an avowedly *cognitivist* project is that the meanings of lexical items compose in ways that suitably mirror the structure of complex concepts. Thus, having identified the meanings of lexical items with instructions to fetch individual concepts, he argues that these instructions compose, forming *complex* instructions, with some functioning as (detachable) components of others. These semantic instructions—what Pietroski calls *Begriffsplans*—are responsible for the assembly of concepts meet two constraints. First, they must be suited to *that specific type* of instruction. While other kinds of human concepts might be assembled by non-linguistic means, *Begriffsplans* can only assemble concepts of a very specific nature (to be spelled out shortly). Second, in keeping with the "mirroring" constraint (my word, not his), the complex concepts that *Begriffsplans* assemble must bear the same part-whole relationships to one another as do the *Begriffsplans* themselves.

Laying out some of the specifics of the *Begriffsplans* that Pietroski posits will put us in a position to better appreciate his views on concepts. The clearest case of this pertains to instructions for *predicate conjunction*. Pietroski takes this to be an absolutely central aspect of linguistic concept assembly, in part because he holds that the kinds of concepts that the human FL is capable of assembling are uniformly *predicative*. In saying this, he means to deny outright that natural languages ("Slangs") allow us to access singular concepts. Such concepts do exist, he thinks, but they *can't* be fetched by *Begriffsplans*. Indeed, he holds that the *only* predicative concepts FL can fetch, and hence assemble, are limited to just the monadic and the quasi-dyadic, with higher adicities receiving a different analysis. These two types of concept correspond to two flavors of predicate conjunction: M-junction and D-junction. Here's how Pietroski characterizes the overall process.

> If biology somehow implements M-junction and D-junction, one can envision a mind with further capacities to (i) use lexical items as devices for accessing simple concepts that can be inputs to these operations, and (ii) combine lexical items in ways that invoke these

> operations. … Suppose that combining two Slang expressions, atomic or complex, is an instruction to send a pair of corresponding concepts to a "joiner" whose outputs can be inputs to further operations of joining. Imagine a mind—call it Joyce—that has some lexical items, each with a long-term address that may be shared by two or more polysemously related concepts. Joyce also has a workspace in which (copies of) two concepts can be either M-joined or D-joined to form a single monadic concept, thereby making room for another concept in the workspace, up to some limit. Joyce can produce and execute instructions like fetch@'cow'; where for each lexical item L, the instruction fetch@L is executed by copying a concept that resides at the long-term address of L into the workspace. Joyce can also produce and execute instructions of the forms M-join[I, I_0] and D-join[I, I_0]; where I and I_0 are also generable instructions. An instance of M-join[I, I_0] is executed by M-joining results of executing I and I_0, leaving the result in the workspace, and likewise for an instance D-join[I, I_0].

Having introduced two basic types of composable *Begriffsplans*—one for fetching concepts like DOG(_) and one for assembling these into complex structures—Pietroski adds four other types of basic semantic operation:

(i) a limited operation of *existential closure*
(ii) a mental analog of *relative clause formation* (weaker than λ-abstraction)
(iii) the *introduction* of concepts like GIVE(_) on the basis of GIVE(*x, y, z*)
(iv) of *thematic* concepts—e.g., AGENT(_), PATIENT(_), RECIPIENT(_)

> [G]iven two monadic concepts, the operation of M-junction yields a third such concept that applies to an entity e if and only if each of the two constituent concepts applies to e. (32) … In short, Slangs let us access and assemble monadic [and some limited dyadic] concepts that can be conjoined, indexed, polarized, and used as bases for a limited kind of abstraction.

We'll look at several of these operations in more detail below, but the following passage contains an initial illustration of the kinds of structures that this system can assemble.

> My claim is not that 'gave a dog a bone' is an instruction to build [just *any*] concept with which one can think about things that gave a dog a bone. That instruction might be executed by building the concept ∃y∃z[GAVE(x, y, z) & BONE(y) & DOG(z)], which has a triadic constituent. My claim is that 'gave a dog a bone' is an instruction for how to build an M-junction like [[GIVE(_)^PAST(_)]^∃[PATIENT(_, _), BONE(_)]]^∃[RECIPIENT (_, _)^DOG(_)], which has only an occasional dyadic concept that has been "sealed in."

This passage usefully contrasts the conceptual structures assembled by FL with the those that are often assumed by linguists—wrongly, by Pietroski's lights—to be available to humans *antecedent* to the development of language.

§2.4.3 Concepts, predicative and sentential

We are now in a position to ask more specific questions about Pietroski's third key notion—viz., that of a concept. As we've already seen, he takes these to be expressions in a compositional language of thought, some of which can be assembled by the semantic module of FL. But, however they might be assembled, they are the representations that allow us to think about the world.

> [C]oncepts have contents that can be described as ways of thinking about things; cf. Evans. A concept that can be used to think about something as a rabbit, whatever that amounts to, has a content that we can gesture at by talking about the concept type RABBIT. An instance of this type is a mental symbol that can be used to think about a rabbit as such, or to

> classify something—perhaps wrongly—as a rabbit; see Fodor. A concept of the type RABBIT-THAT-RAN, which can be used think about something as a rabbit that ran, is presumably a complex mental symbol whose constituents include an instance of RABBIT. A thought can be described as a sentential concept that lets us think about (some portion of) the universe as being a certain way. Thoughts of the type A-RABBIT-RAN can be used to think about the world as being such that a rabbit ran. (4)

As the remarks at the end of this passage indicate, Pietroski takes thoughts to be a *special kind of concept*—namely, a *sentential* concept. This is important to highlight, in view of its relation to a broader point about sentential *meanings*.

Pietroski is skeptical that "Slangs generate sentences as such." The traditional notion of a sentence, as a unity of a subject and a predicate, has been roundly abandoned in contemporary linguistics. While the notions of "subject" and "sentence" have a place in subject-predicate conceptions of *thought*, Pietroski points out that they "may have no stable place" in contemporary scientific grammars (114).

> Linguists have replaced "S" with many phrase-like projections of functional items that include tense and agreement morphemes, along with various complementizers. This raises questions about what sentences are, and whether any grammatical notion corresponds to the notion of a truth-evaluable thought. But theories of grammatical structure—and to that extent, theories of the expressions that Slangs generate—have been improved by *not* positing a special category of sentence. So while such a category often plays a special role in the stipulations regarding invented formal languages, grammatical structure may be independent of any notion of sentence. (61)

Accordingly, Pietroski suspects that talk of "grammatical subjects" is just a roundabout way of "saying that tensed clauses have a 'left edge constituent' that somehow makes them complete sentences—whatever that amounts to—as opposed to mere phrases like 'telephoned Bingley'" (87). Rather than clarifying the notion of a "complete sentence," Pietroski points out that talk of grammatical subjects *presupposes* it.

How, then, to characterize sentences? Naturally, Pietroski does *not* appeal to a distinction between sentential truth conditions and subsentential satisfaction conditions. Instead, he develops a novel version of predicativism, according to which *all* of the concepts assembled by *Begriffsplans* are predicative, in the sense that they all have a classificatory function. This includes concepts that are fetched by linguistic expressions like 'Jessica', 'David Pereplyotchik', and 'Reykjavík'. (Yes, *the* Reykjavík.)

So far, the view on the table is a version of the familiar predicativist position that was introduced by Quine (1970), defended by Burge (1973), and reanimated in contemporary discussions by the work of Delia Graff Fara (2005). Pietroski goes on, however, to make a quite novel claim—namely, that the concepts assembled by *sentence*-sized Slang expressions are *also* predicative.

The idea is that familiar subsentential predicates are assembled, largely via predicate conjunction, and then a new mental operation (⇑ or ⇓) converts the results into a sentential predicate—what Pietroski calls a "polarized concept". Here is how he defines these: "Given any concept M, applying the operation ⇑ yields a polarized concept, ⇑M, that applies to each thing if M applies to something" (30). For instance, if RABBIT applies to something, then ⇑RABBIT applies to each thing and ⇓RABBIT applies to no-thing. We will return to this topic in §3, when we compare this proposal with the inferentialist account of sentence meaning.

Recall that semantic instructions (*Begriffsplans*) have "mechanical execution conditions". Because Pietroski takes *Begriffsplans* to *be* linguistic meanings, it follows for him that that "meanings satisfy demanding compositionality constraints." Such constraints, he argues, permit the assembly of concepts that are better suited for their role in language use than for the epistemic role of "fitting the world". This important upshot of Pietroski's view bears on his rejection of both Davidson's extensional semantics and Lewis's unrestricted type-theoretic approach to natural language (§2.1). For, although he leaves it open that we might build truth-evaluable thoughts as a *side-effect* of language processing, he denies that "meanings are instructions for how to build concepts that exhibit classical semantic properties" (115). Likewise, he suspects that "most natural concepts [do not] have extensions; cp. Travis… if only because of vagueness; cp. Sainsbury" (9). Hence, the *Begriffsplans* that Pietroski identifies with meanings "make no reference to the things we usually think and talk about" (115). If correct, this conclusion is just one more nail in the coffin of the extensionalist project.

§2.5 Pietroski on Fregean thoughts and concepts

Common to both Pietroski and Brandom is a deep engagement with the work of Frege. However, as we'll see presently, the lessons that Pietroski draws from Frege are *not* those that one might expect. In particular, the formal device that he takes over from Frege's semantics is not that of function application, as is common; rather, he emphasizes Frege's immensely useful notion of concept *invention*—something you don't hear much about in discussions of Frege, at least amongst linguists.

As noted earlier, Pietroski holds that are multiple languages of thought—i.e., distinct formats of concept application. In his discussions of Frege, he advances the hypothesis that there are, in fact, at least two such languages. The first one, in order of evolutionary history, may well have a Fregean semantics and include expressions of type <t>. The second one, which only came in with the evolution of natural language, consists of concepts that were *invented*, or *introduced*, in a Fregean sense, on the basis of the older ones.

> [N]atural sentences of type <t> may belong to languages of thought that are phylogenetically older than Slangs. Expressions of these newer languages may be used to build complex monadic concepts, perhaps including some special cases that are closely related to natural thoughts of type <t>. In which case, the very idea of a truth-conditional semantics for a human language may be fundamentally misguided. (114)

Because Pietroski treats the new type of concept as being invariably predicative—i.e., functioning semantically to classify things into *categories*, not to denote them individually—he calls such concepts "categorical". The older type of concept, which participates in thoughts of type <t>, includes singular denoting concepts and predicates of any adicity. On account of their semantic function of relating items to each other, Pietroski calls such concepts (and the thoughts they participate in) "relational".

Though I see its significance, I'm not, myself, a huge fan of the 'categorical'/'relational' *terminology*. Adverting to their historical roles, rather than their internal logic, I'll call these languages Olde Mentalese and New Mentalese for the remainder of the discussion. Here's how Pietroski casts the theoretical relations between them.

> Frege assumed that we naturally think and talk in a subject-predicate format, and that we need help—[e.g.] his invented *Begriffsschrift*—in order to use our rudimentary capacities for relational thought in systematic ways… The idea was that a thought content can be "dimly grasped," in some natural way, and then re-presented in a more logically

> perspicuous format that highlights inferential relations to other contents… I think this is basically right: our categorical thoughts are governed by a natural logic that lets us appreciate certain implication relations among predicates; but our relational concepts are related in less systematic ways. We use relational concepts in natural modes of thought. (95-6)

The distinction between Olde Mentalese and New Mentalese allows Pietroski clarify his perspective, contrasting it with Frege's. Here, too, it's instructive to quote at length.

> Frege introduced higher-order polyadic analogs of monadic concepts. In this respect, my project is the converse of his. Frege invented logically interesting concepts, and he viewed monadicity as a kind of relation to truth, as part of a project in logic that prescinds from many details of human psychology. I think humans naturally use concepts of various adicities to introduce logically boring predicative analogs. But I adopt Frege's idea that available concepts can be used to introduce formally new ones, and that this can be useful for certain derivational purposes. Frege "unpacked" monadic concepts like NUMBER(_), in ways that let him exploit the power of his sophisticated polyadic logic to derive arithmetic axioms from (HP). I am suggesting that Slangs let us use antecedently available concepts—many of which are polyadic—to introduce concepts like CHASE(_) and GIVE(_), which can be combined in simple ways that allow for simple inferences like conjunction reduction. But the big idea, which I am applying to the study of Slangs, is Fregean: languages are not mere tools for expressing available concepts; they can be used to introduce formally new concepts that are useful given certain computational capacities and limitations. This is why I have dwelt so long on Frege's project. For while the idea of concept introduction was important for Frege, it is not the aspect of his work that semanticists typically draw on.

The gory details of Frege's technical devices for concept introduction are, mercifully, beyond our present needs; only a few key points are relevant. One is that introducing concepts need *not* be seen on the model of explicit definition. Rather, Pietroski highlights Frege's proposal for a *second* way of introducing concepts—viz., by *inventing* them. Similarly, although *analyzing* a concept has often been seen as breaking it down into its more basic definitional constituents, Pietroski joins Fodor (1970) in rejecting the idea that lexicalized concepts will generally admit of such analytic definitions. Nevertheless, there is an alternative way of analyzing concepts, which Pietroski characterizes as "a *creative* activity" (emphasis mine).

> Given a very fine-grained notion of content, or thought-equivalence, analysis may not be possible. But Frege employed at least two notions of content: one based on his notion of sense (*Sinn*), and another according to which thoughts are equivalent if each follows from the other. Given the latter notion, or Lewis's characterization of contents as sets of logically possible worlds, one can say that our current representations are not yet perspicuous. We can use our concepts to ask questions that lead us to reformulate the questions in ways that allow for interesting answers. From this perspective, analysis can be a creative activity whose aim is not to depict our current representations…

It's in virtue of our ability to invent new concepts that we, *qua* humans endowed with a specific FL, have invented the monadic and quasi-dyadic concepts that arise only for language use. This includes not only monadic event-predicates like GIVE(_), invented on the basis of the older triadic concept GIVE(x, y, z), but also—importantly for Pietroski's purposes, though not ours—*thematic* concepts such as AGENT(_), PATIENT(_), and RECIPIENT(_).

§2.6 Summary

The generativist methodology that animates Pietroski's inquiry leads him to a number of strikingly original claims about concepts and a detailed theory of meanings. Treating the latter in a resolutely naturalist fashion, he maintains that their theoretical role is to mediate between pronunciations and concepts—i.e., to effect the psychological operations that constitute the interface between language (FL) and the "conceptual-intentional system" (to use Chomsky's coinage). Although meanings facilitate the *assembly* of concepts, which *have* intentional contents, Pietroski holds that meanings are neither concepts *nor* their contents.

On this view, the relation between truth and conceptual/intentional content is "quite complicated and orthogonal to the central issues concerning how meanings compose" (115). This, among the many other reasons surveyed above, leads Pietroski to abandon Davidson's project of extensional truth-conditional semantics. Moreover, the goal of *explaining* our access to a productive hierarchy of concepts, rather than merely *stipulating* it, underlies his rejection of the type-theoretic approach championed by Lewis (1970)—one of the many disagreements that we'll look at in the next section.

The semantic theory that satisfies Pietroski's methodological commitments—as well as the compositionality constraints that he argues follow from it—treats meanings as composable instructions for concept assembly. The instructions are "composable" in the sense that their basic constituents—namely, *fetch@* and *join[I, I']*—can enter into part-whole relations to one another. Moreover, as noted earlier, the larger structures they compose will, in a definite sense, *mirror* those of the concepts that the instructions assemble.

Having furnished empirical evidence for the idea that these "*Begriffsplans*" reduce largely to two flavors of *predicate conjunction*, Pietroski adopts a strong version of predicativism, according to which *all* of the concepts that natural language allows us to access and assemble are predicative. This includes not only the concepts fetched by linguistic expressions that have *traditionally* been classed as predicates, but also those that have generally been seen as differing in some important respect—including singular terms and, more strikingly, even *sentences*. The conceptual predicates that meanings allow us to access and assemble thus all either monadic, dyadic (in a restricted sense), or "polarized", where the latter kind is assembled by sentence-like linguistic expressions, using specialized mental operations, ⇑ and ⇓, to "polarize" concepts. Importantly, the resulting conceptual structures are not necessarily ones that best "fit the world", and they're not even the only ones we can deploy in thought. But, if Pietroski is correct, the they *are* the only ones that FL can assemble.

Denying that the concepts involved in language use have denotational properties and relational structures (of arbitrary adicity) leaves open whether *other* concepts might have these features. As we saw, Pietroski hypothesizes that there *are* in fact such concepts, and that they belong to a phylogenetically older language of thought than the one FL allows us to access—what I've dubbed 'Olde Mentalese'. Olde thoughts might have a subject-predicate form, a Fregean semantics, and belong to the semantic type <t>.

Pietroski goes on to make novel use of Frege's notion of concept *invention* in explaining the (non-definitional) mental introduction of *new* concepts on the basis of the Olde ones—specifically, the ones that FL allows us to access/assemble (New Mentalese). This psychological process, he argues, serves to introduce GIVE(_) on the basis of GIVE(*x, y, z*), as well as novel *thematic* concepts such as AGENT(_), PATIENT(_). These, in turn, participate in building polarized sentential concepts, such as ⇑RABBIT, which "applies to each thing if RABBIT applies to something". In the course of assembling such concepts, it may happen—*but only as a side-effect* (fortuitous or otherwise)—that we *also* token thoughts of Olde Mentalese. But the details of how Olde Mentalese thoughts functions are, Pietroski rightly holds, beyond the scope of a naturalistic semantic inquiry into human *language*.

§3. Prospects for Ecumenicism

Introduction

We've now surveyed the core commitments of two large-scale theoretical frameworks in the philosophy of language and seen some of the ways in which they play out in the realm of semantics, including in detailed analyses of various linguistic constructions. It may appear that the two views are so different in substance and overall methodology that a conversation between the two is unlikely to bear much fruit. In fact, I suspect this is a large part of why so few conversations of this kind ever take place. In the present section, I argue for the contrary perspective, outlining an ecumenical approach that seeks to integrate the two in a variety of ways. In surveying what I take to be significant points of convergence—which then serve as background for constraining residual disputes—I rebuff various superficial objections to the possibility of integration. In each case, I show how the theoretical differences that they point to can be reconciled without doing much (if any) violence to either view.

§3.1 Truth, reference, and other non-explanatory notions

One obvious shared commitment between Brandom and Pietroski—indeed, the one that most clearly motivates the present enterprise—is their common rejection truth-conditional semantics. We've seen a lot about this already, but let's review the key points and add some new ones.

Pietroski surveys a battery of arguments against Davidson's proposal, including its more recent incarnations in possible-worlds semantics. These include troubles with (i) empty names, (ii) co-extensive but non-synonymous expressions, (iii) polysemy, (iv) compositionality, (v) liar sentences, and (vi) event descriptions (*inter alia*). Brandom's skepticism is more foundational. On his view, an explanatory strategy that takes truth and reference—conceived of as "naturalizable" word-world relations—as fundamental semantic relations will require a metasemantics that is, at best, optional, at worst, incoherent, and, at present, non-existent. Although he doesn't pretend to have supplied a knock-down argument against it, the flaws he identifies in the various attempts to work out this strategy strike me as fatal. Coupled with his development of a powerful alternative—a large-scale framework constructed from the top down, with pragmatics taking an unconventional leading role—as well as his well-motivated treatment of the notions of truth and reference, Brandom deals a serious blow to the mainstream approach of subordinating pragmatics to semantics.

Brandom's *main* reason for pursuing a normative metasemantics is the one we saw at the outset—i.e., the inability of a purely descriptive ("naturalistic") account to capture the normative notion of (in)correct rule-following. But this is not his *only* argument, and it's worth taking a moment to spell out what strikes me as a potentially more powerful consideration—one that, in making fewer assumptions, can appeal to theorists of a broader stripe.

All extant attempts at "naturalizing" meaning, content, representation, and the like, have in common their insistence on employing in only *alethic* modal notions in their analyses. These include dispositions (Quine), causation (Stampe), causal covariance (Locke), natural law (Dretske), nomic possibility (e.g., Fodor's "asymmetric dependence"), and even appeals to non-normative teleology (Cummins and, independently, Millikan). Brandom points out that, even if these *could* account for ordinary empirical concepts, it's not at all clear how they might be extended to the very concepts that appear in the analyses—i.e., the *alethic modal* notions just listed (among others).

While it's possible to envision, if only dimly, how something in a person (or a brain) can causally co-vary with—or bear nomological relations to—water, mountains, and even crumpled shirts, there's simply no naturalistic model for envisioning the relation between, on the one hand, the words 'possible', 'disposition', or 'asymmetric dependence', and, on the other hand, any particular set of things, events, or phenomena out in the natural world. The same is arguably true for logical, mathematical, semantic, and deontic vocabulary, as previously noted. (Recall the fonts, functions, and fears from §2.3.) Indeed, the metasemantics doesn't seem to get us much farther than 'rock' and 'stick'. And the experts seem to have given up, since the late 1990s, on the hard at work of bringing 'fail' and 'decisively' into the fold. One is well-advised not to bank on any striking new developments in this area, unless, of course, something dramatic happens in the surrounding domains of inquiry. (My money, for what it's worth, is on the AI people.)

By contrast, a metasemantics that makes explicit and essential use of *normative* terms—paradigmatically, deontic modals—is ultimately able to "eat its own tail", to use Brandom's imagery, by shoring up a principled account of *those very notions*. We've already seen a bit about how Brandom treats normative expressions. According to him, they all serve the function, characteristic of logical vocabulary more generally, of expressing (making explicit) one's commitments to the propriety of an inference or a plan of practical action. Brandom (2008) offers, in addition, an account of *alethic* modal vocabulary (recall the safety measures for gasoline wicks in §1.2), as well as a detailed formal analysis of its many important relations, both semantic and pragmatic, to the deontic variety. In this way, his account can claim a major advantage over virtually any *conceivable* attempt at a naturalistic alternative. And, again, the force of this argument does *not* depend on a prior assumption about the normativity of meaning. This functions here not as a premise, but as a conclusion.

§3.2 Naturalism

Residual worries about adopting a normative metasemantics will doubtless trouble self-avowed naturalists (including former versions of the present author), who tend to have a constitutional aversion to trafficking in normative considerations. But this, too, should be tempered—or so I'll now argue. The concern is, to my mind, largely dampened by the fact that Brandom's norms are in no way "spooky" (despite drawing heavily on Kant), but, rather, grounded directly in social practices. Such practices consist of activities that are themselves rooted in each creature's *practical stance* of reciprocal authority and responsibility to all others in its community.[18] Such stances are overtly enacted and then, over time, socially instituted in a *wholly non-miraculous* fashion, by *natural* creatures.

Moreover, the resulting discursive/conceptual activities are open to assessment in respect of truth, accuracy, correctness, and overall fidelity to "the facts on the ground" (as the assessor sees them). Indeed, the project of normative pragmatics is so obviously *not super*natural that it's not clear why the self-avowed "naturalist" should be at all worried. Even less clear is why *anyone*

[18] There's been confusion on this point, caused largely, I think, by Brandom's uncharacteristically ill-chosen terminology. He gives the label "practical attitudes" to what I've here (re-)described as "embodied practical stances" on the part of a creature toward its community members. The use of the term 'attitude' has predictably conjured in the minds of some critics the notion of a propositional attitude—something that already bears a distinctively conceptual/intentional content. Plainly, this would render Brandom's account viciously circular, as he is aiming to explicate the notion of propositional attitude content *in terms of* (what he calls) practical attitudes. If the latter already have intentional contents, then there's obviously no difficulty in spelling out other semantic/intentional notions downstream. Equally obviously, there would be no theoretical interest in doing so.

should get to dictate the terms of legitimate inquiry *a priori*. Why, after all, should our metasemantic theorizing *not* make any use of the perfectly familiar and logically well-behaved *deontic* modal notions. Indeed, why even a *lesser* use of them than their alethic counterparts? What's so special about *alethic* modality, anyway? Nothing much, so far as I can see.

Let me dwell on this point, for it seems to me that the knee-jerk resistance to normative theorizing is deeply ingrained in the naturalist's mind. (I should know!) Pressing back against what now strikes me as an irrational prejudice, I exhort philosophers to actively discourage it, *whatever* the fate of Brandom's philosophical project—or, for that matter, mine—turns out to be. Given our daily immersion in social norms and institutions, it's frankly puzzling that so many theorists have allergic reactions to a deontic treatment of language. Norms are *not* puzzling. They are all around us, every moment of our lives. They permeate every social interaction we have and they are the subject of most of our thoughts, *all* of our plans, and our very conceptions of our own identities as free, responsible agents.

Moreover, with respect to *linguistic* norms in particular, there are (so far) no *obvious* examples in the natural world of linguistic abilities arising in creatures *outside* of a relatively elaborate social context. Indeed, even intelligent artifacts wouldn't count, if we ever made one, for they'd be related to the human community of happy roboticists in an obviously relevant way. So it's not at all clear—not to me, at least—why *this* aspect of naturalism should constrain our inquiries into language and mentality. Obviously, naturalism has many *other* appealing features, but this doesn't seem to be one of them.

The deafening silence from classical naturalists on this point has led some, e.g., Price (2010, 2013), to endorse Brandom's normative inferentialist project and to embed it into a larger philosophical framework that eschews notions of correspondence, reference, "the representation relation", etc. altogether. (No "mirrors", he enjoins, using Rorty's metaphor.) Price applies Brandom's expressivist account of logical vocabulary to *all* of human language. The resulting "*global* expressivism" is a key commitment of the novel brand of naturalism that he recommends to our attention—one that I find deeply compelling.

What I've been describing as the "traditional" or "classical" naturalist view—i.e., the received view among *soi-disant* naturalists in the literature from Fodor onwards—maintains that we should draw on the tools, models, and concepts of natural science in characterizing atomistic word-object or sentence-fact relations—paradigmatically, reference and correspondence. On this picture, "the world" is seen through the lens of natural science—a metaphysical framework that has plenty of room for protons, genes, and brains, but stubbornly refuses to accommodate responsibilities, entitlements, and the like—including, remarkably, *persons* (at least not in the fullest sense of that word; cf. Sellars, 1963). The "objects" to which language relates us are thus limited by naturalist maxims to only the "natural" ones, whatever *those* are. For this reason, Price calls this view *object* naturalism.

The alternative that Price puts on offer is *subject* naturalism. This view retains a healthy and well-deserved respect for the deliverances of natural science, but refuses to go along with the *philosophical* fiction of "naturalizable" reference and correspondence relations. Rather, our naturalistic urges should be directed, Price argues, toward concept- and language-using *subjects*—i.e., the creatures who acquire, produce, and consume languages, as just *one* tool in a larger biological-*cum*-social enterprise of maintaining homeostasis in the species. Paradigmatically, such creatures are human persons, but any other naturally social creature can in principle be studied in this fashion.

What's striking, to my mind, is how similar all of this sounds to the methodological aims of theorists like Chomsky and Pietroski. Although both call themselves naturalists, each has made determined efforts to debunk the idea that word-world relations are relevant to an empirical study of the human language faculty. Nor does either theorist harbor the ambition—characteristic of the "classical" naturalists mentioned earlier—to *reduce* intentionality, either by analysis or metaphysically, to some alethic modal base. Here's Pietroski on the issue:

> One can raise further questions about the sense(s) in which *Begriffsplans* are intentional, and which philosophical projects would be forfeited by appealing to unreduced capacities to generate and execute the instructions posited here; cp. Dummett. But my task is not to reduce linguistic meaning to some nonintentional basis. It's hard enough to say what types of concepts can be fetched via lexical items, and which modes of composition can be invoked by complex Slang expressions.

This is another point of convergence with Brandom's pragmatism, which likewise renounces the reductive aims of the classical naturalist project.

Of course, the mere fact that Pietroski declines to take up the issue in *CM* doesn't mean he has no dog in the fight elsewhere. (I don't, myself, know.) By the same token, although it's true that Brandom doesn't aim to reduce intentional notions to some construct of natural science, it doesn't follow, and isn't true, that he has *no* reductive ambitions at all. To the contrary, his normative inferentialism is designed precisely to reduce intentionality to *something* nonintentional, which, in his case, happens to be the normative. This is why normative pragmatics serves for him as a *meta*semantics, in the fullest sense of the word. The 'meta' indicates not only that what's on offer is a "theory of meaning"—rather than a first-order "meaning theory", to use Dummett's distinction; more importantly, it connotes that the semantics is herein *subordinated* to (i.e., must "answer to", in an explanatory sense) the social norms that are the centerpiece of the pragmatics.

§3.3 Referential purport

In keeping with his commitment to the methodological tenets of individualism and internalism, Pietroski applies many of the points scouted above to *conceptual thought*.

> [S]ome readers may be unfamiliar or uncomfortable with talk of using representations to think about things in certain ways. So some clarification, regarding aboutness and ways, is in order. … The relevant notion of thinking about is *intentional*. We can think about unicorns, even though there are none. One can posit unicorns, wonder what they like to eat, diagnose various observations as symptoms of unicorns, etc. Similarly, one can hypothesize that some planet passes between Mercury and the sun, call this alleged planet 'Vulcan', and then think about Vulcan—to estimate its mass, or wonder whether it is habitable—much as one might posit a planet Neptune that lies beyond Uranus. An episode of thinking about something can be such that for each thing, the episode isn't one of thinking about that thing. … [I]n the course of proving that there is no greatest prime, it seems an ideal thinker can at least briefly entertain a thought with a singular component that purports to indicate the greatest prime. … Paradoxes like Russell's remind us that even idealized symbols can fail to make the intended contact with reality, at least if one is not very careful about the stipulations used to specify interpretations for the symbols.

I want to highlight a key point here: Pietroski is presupposing about concepts *not* that they *succeed* in referring—though he allows that some of them do—but that even *empty* concepts have intentional contents. These latter plainly *cannot* be accounted for by positing straightforward metaphysical relations between words, on the one hand, and bits of the world, on the other.

This emphasis on intentionality in the sense of referential *purport* is crucial to Brandom's project as well. Rather than setting out to explain *successful* reference/denotation, as the paradigms of perceptual and demonstrative reference have led many theorists to do (§2.3), Brandom sees it as necessary to *first* explain how a creature can so much as *purport* to refer to one thing rather than another, and only later to furnish an account of what counts as *success* in this regard. Brandom is not alone in adopting this strategy. In the Gricean tradition, the homologous project is cast in terms of the *intentional design* of communicative acts—in particular, a speaker's *intentions* to refer, denote, predicate, or to speak truly of something. But whether one uses the idioms of purport, design, or intention, the key point is that the phenomenon under discussion does *not* involve unique, naturalizable, or semantically-relevant mind-world relations.

That leaves wide open issues about the interface between language and reality, let alone larger questions of metaphysics and epistemology. While Pietroski stays largely neutral on such topics in *CM*—again, justifiably so—Brandom's account makes quite definite commitments in these arenas. Nevertheless, there is one place where the two quite clearly converge, and that is with respect to their treatments of *de dicto* and *de re* attitude ascriptions. Let's take a look at that.

§3.4 De dicto *and* de re *constructions*

In granting reasonable concessions to philosophers who stress the importance of mind-world relations for our theories of intentionality, meaning, concepts, and the like, Pietroski makes the following remarks:

> I grant that 'think about' can also be used to talk about a relation that thinkers can bear to entities/stuff to which concepts can apply; see, e.g., Burge. In this "*de re*" sense, one can think about bosons and dark matter only if the world includes bosons and dark matter, about which one can think. Any episode of thinking *de re* about Hesperus is an episode of thinking *de re* about Phosphorus. This extensional/externalistic notion has utility, in ordinary conversations and perhaps in cognitive science, when the task is to describe representations of a shared environment for an audience who may represent that environment differently. But however this notion is related to words like 'think' and 'about', many animals have concepts that let them think about things in ways that are individuated intentionally. We animals may also have representations that are more heavily anchored in reality; see, e.g., Pylyshyn. But a lot of thought is intentional, however we describe it. (80)

It's important to see that there are two distinct strands of thought here. One is about how *some* representations—again, paradigmatically those involved in perceptual or demonstrative reference—are "more heavily anchored in the world" than others. This might *seem* to put Pietroski's view at odds with Brandom's inferentialism, given how small an explanatory role the latter gives to such "anchoring" relations. But this is worry is spurious.

Brandom's account of perceptual commitments and default entitlements (*MIE*, ch. 4), as well as his (largely independent) account of demonstrative reference and "object-involving" thoughts, are fully compatible with—indeed, positively *require*—the existence of reliable nomic or causal relations between perceptible objects in the world and the perceptual mechanisms of a creature. There are, to be sure, heated debates about how exactly all of that works—e.g., whether the percepts should be seen as having the function of bare demonstratives (Fodor and Pylyshyn, 2015), noun phrases (Burge, 2010), or inferentially integrated singular thoughts (Brandom, 1994). But disagreements on these points are far downstream, theoretically, from the broadly methodological commitments that I want to highlight here. It's these that are the subject of the *second* strand of thought that I think we should distinguish here.

When Pietroski speaks of a "'*de re*' sense" in which one can think about or ascribe thoughts, he is at once talking about a certain kind of *thought*—the "object-involving" kind discussed above—but also, separately, about a certain kind of thought *ascription*. The latter, he says, "has utility, in ordinary conversations and perhaps in cognitive science, when the task is to describe representations of a shared environment for an audience who may represent that environment differently." Note that this circumscribes the function of *de re* ascriptions to what I think of as "the navigation of perspectival divides". More prosaically, *de re* constructions allow language users to describe the environment in which a creature is deploying its concepts—as viewed *by the describer* (and often her audience)—while *de dicto* constructions function to ascribe *the concepts so deployed*. Here, then, we see another major point of convergence between Pietroski and Brandom. The latter provides an inferentialist analysis of *de dicto* and *de re* ascriptions, according which they perform precisely the function that Pietroski's remarks indicate.

According to Brandom, *de dicto* ascriptions make explicit the commitments of the creature being described—not those of the ascriber, who may be either ignorant on the matter or hold commitments directly contrary to those ascribed. A speaker might say, "Dan thinks the greatest philosopher of language was Quine," without having any commitments one way or the other about whether philosophers exist, or about whether Quine was one of the greats. Indeed, the speaker might think that philosophers are bits of tofu and that Quine is a particularly flavorful brand. None of that would matter with regard to the speaker's entitlement to a *de dicto* claim about what *Dan* said (assuming, of course, that the speaker is not identical with Dan himself; see fn. 7).

By contrast, had the speaker employed a construction that functions in the *de re* sense—e.g., the awkward 'of'-constructions that we've inherited from Quine—then their own commitments would have come into play, with questions arising (at least in potentia) about their *entitlements* to those commitments. For instance, had the speaker's ascription been (1), then their own commitments regarding the existence of philosophers, the list of the greats, and the possibility of philosophical tofu would have become immediately relevant.

(1) Regarding the greatest philosopher of language, Dan thinks that *he* was a piece of tofu!

Turning to *sub*sentential cases, Brandom points out that such shifts in perspectival commitments can be indicated by operators such as "classified as", "described as", "conceptualized as", and (importantly) "referred to as". For instance, in saying "Jamal classified some food *as rabbit*," a speaker, Juanita, purports to indicate some food—the *de re* component of the ascription—and then says what concept Jamal applied to it (viz., RABBIT). The word 'as' marks the onset of the *de dicto* component, ensuring that Juanita does not commit *herself* to the correctness of Jamal's classification. (Perhaps she knows that that the stuff on the plate is tofu.)

Some theorists, having noted that RABBIT is the *only* concept that Juanita ascribes to Jamal, go on conjecture that Jamal can deploy this subsentential concept, *all by itself*, in classifying something as rabbit. Indeed, bewitched by surface grammar, some fail to notice the plain distinction between *what Juanita is doing*—i.e., describing (noncommittally) one *aspect* of Jamal's perspectival commitments—and *what Jamal is thinking*. A mistaken conflation of these two phenomena is what gives rise, I suspect, to the widespread illusion that we can take as it *as a* *datum* that each of us has ability to think of, classify, or refer to something, by deploying *just one subsentential concept* (or linguistic expression). I'll argue later, following Brandom, that this idea is doubtful; *prima facie*, one can neither classify nor think about something without tokening *complete thoughts*. For instance, the case described, Jamal's classificatory act requires tokening the *complete* thought, CCC IS-RABBIT, where CCC is whatever concept he uses in thinking about the food on the plate.

§3.5 Interpersonal similarity *of meaning and content*

Many theorists hold, for reasons that are ultimately unpersuasive, that communication is a matter of passing a message, idea, meaning, or thought from speaker to hearer. As with throwing a frisbee, a successful case in one in which the item sent by the speaker is the very same as the one received by the hearer—if only in the ideal. But, after the mid-century work of Quine, Kuhn, Sellars, and many others who developed broadly holist ideas (e.g., Churchland, 1979), it's hard to see this picture as anything but optional. Brandom's account of communication is one of the many that rejects it outright, modeling communication instead on activities like tango dancing, where partners has to give and take "in complementary fashion"; book-keeping, where each participant "keeps separate books" regarding her own commitments and those of other participants; and baseball games, where a common "scoreboard" shows what complex normative statuses each participant bears to each of the others. (Many have remarked that this is a distinctively *American* philosophy.)

Metaphors aside, Brandom's inferentialism carries an explicit commitment to *holism* about meaning and content. Many follow Fodor and Lepore (1992) in seeing this as the root of several major problems for his view. But the objections that Fodor and Lepore press are virtually all rooted in implausibly strong assumptions about the necessity of meaning/content *identity*—rather than mere similarity—for various philosophical inquiries. These include the projects of adequately characterizing successful communication, interpersonal disagreement, and rational belief revision. It seems to me that Brandom's accounts of these things are perfectly fine as they stand, but fans of holism seem to have gone scarce in recent years, and Fodorian views about meaning/content identity have arguably become the *received* views in the field. In yet another clear case of both his iconoclasm and his significant convergence with Brandom, Pietroski likewise rejects the identity-of-meanings picture, though on grounds that are independent of any holist commitment. Targeting first the extensional account of meaning-identity as co-extensiveness, Pietroski points out that

> speakers can connect the pronunciation of 'pen' to the same meaning without determining a set of pens that they are talking about. If each speaker uses a polysemous word that lets her access a concept of writing implements, as opposed to animal enclosures, they resolve the *homophony* the same way and thereby avoid blatant miscommunication. In this sense, they connect a common pronunciation with a common meaning. But it doesn't follow that any speaker used a word that has an extension…

Later, Pietroski counsels us—wisely, in my view—to give up the whole idea that "successful communication requires speakers [to] use the same meanings/concepts" (33), regardless of the theoretical framework in which this idea is couched (extensional or otherwise). He views it as a mere *idealization* that "members of a community have acquired the *same* language—or that they use the same words, or words that have the same meanings." Despite Brandom's focus on social norms, shared commitments, and the like, what Pietroski says here is entirely in line with his view. This will take some spelling out, which I undertake in §4. For now, let me emphasize that, if correct, then Pietroski's take on this issue would severely undermine the (already fairly flimsy) arguments against meaning holism.

§3.5 The Pragmatics of Assessment

We've seen that Brandom places great stress on the notion of assertion, and that he sees this as something that we should characterize in normative terms. Given Pietroski's naturalist commitments, one might think that he disagrees. But this overlooks the key point that I will go on to make in the remainder of this discussion—namely that the theoretical aims of Pietroski's semantic theory are so starkly different from those that animate Brandom's inquiry that their common use of folk terms like 'meaning' should not bewitch us into thinking that they're talking about the same phenomenon.

Pietroski's target throughout *CM* is not communication, but the psychological mechanisms that make it so much as possible. When he does provide hints of his broader views about communication, what he says is entirely of a piece with Brandom's normative pragmatism.

> My own view is that truth and falsity are properties of certain evaluative actions—e.g., episodes of assertion or endorsement—and the corresponding propositional contents that are often described with the polysemous word 'thought', as in 'the thought that snow is white' or 'the thought constructed by executing that instruction'; cp. Strawson.

As with all theory-laden terms, including (especially in *this* context) 'thought', 'meaning', and 'concept', we should always remember that the aims and presuppositions of the inquiry are far more important to keep in view than the pronunciation of the jargon. Pietroski makes this point in the following passage.

> Let's not argue about nomenclature. One can use 'concept' more permissively than I do, perhaps to include images or other representations that are not composable constituents of thoughts. One can also use 'concept' less permissively, perhaps to exclude representations that fail to meet certain normative constraints. Or one might reserve the term for certain contents, or ways of thinking about things, as opposed to symbols that have or represent contents. But I want to talk about a kind of physically instantiated composition that is important for cognitive science, along with a correspondingly demanding though non-normative notion of constituent, without denying that contents and nonconceptual representations are also important.

We will go on in §4 to compare Brandom's and Pietroski's notions of "constituent," where it will become important that Pietroski's aims is to provide a *non*-normative account of constituency. For now, consider how this plays out with respect to Pietroski's distinction between Olde and New Mentalese. You'll recall that Pietroski's account has children starting off with Olde Mentalese concepts that *fail* to meet the conditions for assembly by FL *Begriffsplans*, and later *inventing* new concepts that are specially tailored for the job. For instance, Pietroski writes, "Ignoring tense for simplicity, we can form concepts equivalent to: GIVE(VENUS, __ , BESSIE); GIVE(VENUS, BESSIE, __); GIVE(__ , BESSIE, VENUS); GIVE(__, BESSIE, __); etc. But children may start with composable concepts that are *less* cognitively productive" (101).

In the remainder of the discussion, I'll lean heavily on the following two key points. First, much of what Pietroski says about the phylogenetically older "Fregean" modes of thought that permit internal substitution is of a piece with the lessons that Brandom likewise draws from Frege. The second key point, which alone serves to resolve many of the apparent conflicts between the two views, is that Pietroski's account is best be viewed as a (partial) theory of the *subpersonal* mechanisms that underlie our norm-tracking abilities, which in turn contribute to making possible the social practices of communication. For instance, Pietroski's account of how a child comes to acquire the concepts that FL can assemble should, if correct, be regarded as laying down empirical constrains on the variety of norm-tracking social practices that are a human child will be able to master at various ages.

If the foregoing claims are correct, then the two theoretical enterprises not only share several key commitments, but they are actually *complementary*, each providing an important piece of the overall puzzle about how best to view language(s). While Brandom's inquiry is neither descriptive nor psychological, his account clearly *presupposes* that there must be *some* descriptively correct account of how any creature's sub-creatural cognitive architecture emerges, whether in early development or in its species' socio-evolutionary ("memetic") history. His claim is simply that such an account can't be the *end* of the story. Rather, he argues that it's the necessary groundwork for a much larger picture of the role of language in social practice.

Likewise, although Pietroski's focus is the psychology of the individual speaker-hearer—assumed to be universal within the species—we've seen that there's room for communal norms in his overall picture. Indeed, passages from *CM* contain explicit remarks about the norms that govern inquiry in the mature sciences—e.g., norms that serve to stabilize referential purport among expert chemists. Such norms, however, would reside largely at the level of *pragmatics*, which is decidedly *not* Pietroski's focus in *CM*, nor a major aspect of Chomsky's own work since the 50s.

Perhaps, then, we can adapt David Marr's famous distinction between "levels of analysis"—if only by crude analogy—by viewing Brandom as articulating a "high-level description" of the conditions that any language user must satisfy in order to count as such. If this suggestion is right, then Pietroski is best seen as telling us about the details of how this high-level description happens to be "implemented" or "realized" in the human case. Whatever the merit of the analogy to Marr's picture, my claim is that there is no inherent conflict between accounts pitched at distinct levels of inquiry. There could only be a substantive dispute if the they shared a common domain. But, as Pietroski's remarks in the following passage make clear, that's just not so in the present case.

> Let me digress, briefly, to note a different response to concepts without *Bedeutungen*. One might adopt a hyper-externalistic understanding of 'ideal', so that no ideal thinker has the concept VULCAN, and no ideal analog of this concept lacks a *Bedeutung*; cp. Evans, McDowell. But then an ideal thinker must not only have her mental house in good order, avoid paradox-inducing claims, and be largely *en rapport* with an environs free of Cartesian demons. In this hyper-externalistic sense, ideal thinkers are immune from certain kinds of errors, perhaps to the point of being unable to think about the same thing in two ways without knowing it; cp. Kripke. One can characterize corresponding notions of concept*, language*, and their cognates. These notions, normative in more than one way, may be of interest for certain purposes. Perhaps inquirers aspire to have languages* whose expressions* have meanings* that are instructions to build concepts* that can be constituents of thoughts*. But my aim, more mundane, is to describe the thoughts/concepts/meanings that ordinary humans enjoy. And I see no reason to believe that the best—or even a good—way to study concepts or meanings is by viewing them as imperfect analogs of their hyper-externalistic counterparts, even if we can and should try to acquire concepts*. Notions like truth and denotation may be required to interpret philosophical talk of thoughts*/concepts*/meanings*. One can also hypothesize that ordinary words combine to form sentences that express thoughts composed of concepts that denote or "are true of " things in the environment. But this hypothesis does not become inevitable as soon as we talk about thoughts/concepts/meanings. On the contrary, natural concepts seem to be mental symbols that can be used to think about things, even when the concepts apply to nothing. And if words like 'give' are used to access naturally introduced concepts like GIVE(_), we must be prepared to discover that words are used to access concepts that fall short of certain philosophical ideals.

§4. Challenges for Ecumenicism

The ground-clearing maneuvers of the previous section put us in a position to explore residual differences between Pietroski and Brandom that threaten to be more substantive. As advertised, I hope to show that the initial appearances are misleading even in some of *these* cases, contrary to received opinion amongst philosophers of language. Still, I'll ultimately admit defeat when we arrive at the very last topic—the banes of predication and singular terms.

§4.1 E-language

One place to look for sources of substantive disagreement is in the vicinity of Chomsky's infamous distinction between E-language and I-language (Chomsky 1986). Chomsky initially drew this distinction with the explicit intention of formulating a key difference between his approach to language and that of Quine, Lewis, and others—what was then arguably the dominant view. Given that Brandom was a student of Lewis's and frequently pays homage to him in his published work, the question arises whether *his* view likewise falls prey to the compelling objections that Chomsky articulated decades ago. We can address this question by focusing on Pietroski's more recent formulation of Chomsky's insights.

Let's begin with the distinction itself. What exactly is an E-language supposed to be?

> Chomsky characterizes I-languages as procedures that connect interpretations with signals. Languages in any other sense are said to be E-languages. Chomsky … isolates a procedural notion and uses 'E-language' as a cover term for anything that can be called a language but isn't an I-language. Thus, E-languages may include certain clusters of behavioral dispositions, heuristics for construing body language, etc. But Chomsky's 'I/E' contrast does connote Church's intensional/extensional contrast, and sets of interpretation-signal pairs are paradigmatic E-languages. Though to repeat, such sets are often defined procedurally.

This negative conception of E-languages ("anything that can be called a language but isn't an I-language") casts such a wide net that it *can't but* apply to Brandom. Certainly, the latter is making no substantive psychological hypotheses about a computational system in the human brain. Nevertheless, as we'll see, much of what Pietroski says in rejecting the study of E-language poses no conflict with Brandom's view. While he argues that David Lewis "wanted to describe our distinctively human language(s)," this is *not* the direction that Brandom takes. His project is to describe language *as such*, not "distinctively *human* languages."

Although my broader claims by no means hang on a particular interpretation of Lewis, it does seem to me that Pietroski's reading of Lewis is not maximally charitable. I say this *not* because I think that his deep disagreements with Lewis somehow insidiously creep into his interpretation. Rather, I'll argue that there's a way of seeing Lewis as engaged in a wholly different project from the one that Pietroski foists on him. Still, whatever the case about Lewis, the project that I have in mind is the one that Brandom *in fact* undertakes—which, in my view, he carries out successfully, even if his teacher did not.

The subsections that follow distinguish three separate points of debate: (i) extensional vs. procedural conceptions of grammar, (ii) a sentence-first vs. word-first approach to semantics, and (iii) individualist vs. social conceptions of language.

§4.1.1 Extensional vs. intensional constraints

In getting more precise about Lewis's particular brand of E-language, Pietroski highlights two points: (i) the metaphysical claim that language(s) are extensionally-specified abstract objects—specifically, sets of meaning-pronunciation pairs—and (ii) Lewis's conception of the process whereby a population comes to *use* such an object in social exchanges.

> Lewis's proposal was that languages like English are sets [functions-in-extension, not procedures] that are related to certain social phenomena via conventions of "truthfulness and trust." He speaks in terms of populations "selecting" certain languages. The suggestion is that using a particular language is like driving on the right: an arbitrarily chosen way of coordinating certain actions.

One of the main things that Pietroski finds problematic here is that treating natural languages as functions-in-extension requires relinquishing the explanatory ambitions of generative linguistics—specifically, the aim of describing the cognitive structure that underlies, or even *constitutes*, human linguistic competence. Lacking an account of the cognitive architecture of FL, or the representational format in which it conducts its business, we have no empirically credible story about how adults accomplish real-time language processing, not to mention how children acquire the creative capacity to produce and consume an indefinite range of novel expressions.

> Lewis says instead that "a grammar, like a language, is a set-theoretic entity which can be described in complete abstraction from human affairs." Chomsky offers a way of locating languages and grammars in nature. Lewis stipulates that each grammar determines the language (i.e., the set of sentences) that it generates, but that languages do not determine grammars. So even if there is a Lewisian grammar for a certain set of Slang sentences, this does not explain how the relevant sentence meanings are related to lexical meanings, or how speakers of the Slang know that the strings in question fail to have certain meanings. At best, a Lewisian grammar indicates how a certain kind of mind might abstract lexical meanings from sentence meanings, given hypotheses about the relevant constituency structures and composition principles; see chapter three. But this doesn't yet tell us anything about how humans connect lexical meanings with pronunciations, or how we can/cannot combine lexical meanings.

This all strikes me as correct. What Lewis *should* have been aiming his inquiry at language *as such*, regardless of which creature is using, acquiring, or "selecting" it. Setting aside Lewis, *Brandom's* target explanandum is "what it is to do the trick," not "how the trick is done by *us*"—or, for that matter, by a dolphin, an AI robot, or a Martian. (I'll suggest a friendly amendment to this point in §4.2). Chomsky, Pietroski, and other generativists, on the other hand, want to know about humans *specifically*. These are *different* projects, to be sure, but they're not thereby *rivals*. Though they may well constrain one another, the relations between them can better be seen in terms of the "levels of analysis" picture that I proposed earlier.

In addressing Lewis's account of how populations "select" a language, Pietroski assumes that the target phenomena are sufficiently similar between his inquiry and Lewis's that the two are not only commensurable with one another, but are actually in direct conflict on various key points.

> Lewis asserts that £ is the language used in a population P "by virtue of the conventions of language prevailing in P;" where conventions, in his sense, are special cases of mutually recognized regularities of action/belief in a population of individuals who can act rationally. He says that a convention of " truth and trustfulness," sustained by " an interest in communication," is what makes a particular set of sentences the language of a given population. However, Lewis offers no evidence that this proposal is correct. And there is an

> obvious alternative: if a population P is small enough to make it plausible to speak of the language used in P, then the members of P will have acquired generative procedures that are very similar. So why think there are "facts about P which objectively select" a shared E-language, and not that members of P have similar I-languages?

The answer that I suspect Lewis would give to Pietroski's last question in this passage is that there is no obvious way of individuating I-languages independently of the "forms of life"—i.e., communal practices—in which any linguistic creature is caught up. That's because a discursive creature's concepts and/or lexical items are constitutively related to the norms that structure those practices. Thus, if meanings really *are* instructions to fetch and assemble concepts, then we won't be able to individuate *meanings* without appealing to such norms *either*. It's not clear to me how Pietroski would (or *could*) respond to this point.

Pietroski's critique of Lewis—particularly, his troublesome notion of "selection"—is carried forward in the following passage. What's new here, though, is that we have a direct quotation from Lewis making precisely the claim that I've been urging on his behalf—namely, that the generative grammarian's target *explananda* are simply *not the ones that* he *seeks to address*.

> According to Lewis, populations select languages by virtue of using sentences in rational ways. So on his view, if language use is not a "rational activity" for young children, they are not "party to conventions of language" or "normal members of a language-using population." [Lewis writes:] "Perhaps language is first acquired and afterward becomes conventional. . . . I am not concerned with the way in which language is acquired, only with the condition of a normal member of a language-using population when he is done acquiring language." I don't know what it is to acquire language in Lewis's sense, or how he would describe whatever creolizers acquire. But even if one wants to focus on "normal" adults, ignoring acquisition may not be a viable option for those who want to find out what Slangs and meanings are. Inquirers don't stipulate how phenomena are related; they investigate.

In the end, I agree with Pietroski that Lewis's talk of populations "selecting" languages is hard to take seriously, especially in light of the Chomskyan alternative. Casting things in Lewis's way plainly does leave the crucial process of "selection" unexplained—at least at the level of psychology, which is arguably where all the *exciting* action is at. Nor can I bring myself to credit the idea that the central aim of formal linguistics should forever be what Lewis said it should be—i.e., the extensional characterization of some abstract set (a moving target, as Pietroski points out, given the constant introduction of novel lexical items). But one can share Pietroski's doubt that there's any meaningful sense in which people "select languages", as well as his broader anti-extensionalism, but *nevertheless maintain* that there *is* a level of theoretical abstraction at which language *is* best viewed as a social phenomenon.

The conventions that govern the phenomenon in question might be "truthfulness and trust", as Lewis thought, or they might be more explicitly normative, as in Brandom's conception of socially-instituted assertional and inferential commitments and entitlements. Likewise, the languages that such conventions institute might be static abstracta, as Lewis seems to have believed, or they might be flexible, dynamic, and highly context-dependent social relations, as on Brandom's account. The latter, moreover, has no theoretical use for the notion of "stable populations" that go around "selecting" various abstract objects—each of which somehow manages to answer to the label 'Spanish' (or whatever), despite their extensional differences.

There is, therefore, no call to reify such strange entities on the normative inferentialist picture. Neither linguistic phyla (e.g., Romance or Germanic) nor local dialects (2019 Boston English) need be real—not in *Lewis's* sense, anyway—in order for Brandom's project to get off the ground.

Indeed, as previously mentioned, the “communities” that Brandom appeals to in his account of norm-governed social practices can be as small as *two creatures*. (Actually, a footnote in *MIE* reveals that Brandom might even be willing to countenance something like an “I-thou relation” coming to be instituted by temporally asymmetric recognition of authority and responsibility relations between *distinct time-slices of one and the same creature*.) Thus, in principle, every new dyadic social interaction can serve to institute novel *local* norms, alongside any that were previously shared. Whatever else Brandom might be accused of doing, then, attempting to individuate and reify stable (let alone timeless) public languages is no part of his brief.

§4.1.2 The primacy of sentences

Having argued against Lewis’s *metaphysical* assumptions, Pietroski goes on to take issue with his *methodological* claim that semantic inquiry should begin by assigning meanings to sentences, rather than to subsentential expressions.

> Lewis goes on to say “ if σ is in the domain of a language £, let us call σ a sentence of £.” But he wasn’t using ‘sentence’ as a technical term for any string to which a language assigns a meaning. Rather, Lewis initially restricted his notion of a meaningful expression to sentences. He later introduces talk of word meanings via talk of grammars.

As we saw in our earlier discussion of Pietroski’s views on sentences, he views this sort of approach as being out of step with contemporary theorizing in generative linguistics.

> Linguists have since replaced “S” with many phrase-like projections of functional items that include tense and agreement morphemes, along with various complementizers. This raises questions about what sentences are, and whether any grammatical notion corresponds to the notion of a truth-evaluable thought. But theories of grammatical structure—and to that extent, theories of the expressions that Slangs generate—have been improved by *not* positing a special category of sentence. So while such a category often plays a special role in the stipulations regarding invented formal languages, grammatical structure may be independent of any notion of sentence.

Here again, we must distinguish Lewis from Brandom. For the latter, the primacy of sentences for is *not* a mere stipulation, nor an irrational fetish for a specific syntactic type. Rather, his principal—and I think quite principled—grounds for isolating sentences at the outset of a normative pragmatic inquiry is that this is the *only* type of expression with which a can make an *assertion*—i.e., an explicit move in an norm-governed *inferential* practice. Brandom’s concern with the normative structure of this “game of giving and asking for reasons” is what motivates his focus on the role that assertions play, both in reasoning—as premises and conclusions—and also in communication. In the latter context, they serve the function of allowing speakers to undertake normative statuses—paradigmatically, commitments and entitlements.

Pietroski’s methodological counsel is to postpone discussion of communication to a later day. As he remarks, specifying the structure of FL is work enough for one lifetime—surely more. But it doesn’t follow from this that a theoretical inquiry with *different* aims must be, in some sense, second-rate, let alone *illegitimate*. Nor is it clear that the two inquiries are even *in competition* with one another over how to best to describe a common subject matter. As noted above, their common talk of “meanings”, “languages”, and the like might tempt one into thinking that the topic under discussion is the same for both theorists. But this would be a mistake. The two theoretical frameworks in which these terms are couched—and relative to which they have their theoretical meanings—are so dramatically different in respect of their explanatory aims that viewing them as even *intending* to refer to the same phenomena is a bit of a stretch.

Unless I'm mistaken, Pietroski himself seems to have fallen into this trap, despite his earlier counsel to avoid such temptations. One clear place where he does so is in the following passage, which is notable for including—now for a *second* time—Lewis's direct protests to being saddled with the views that Pietroski nevertheless goes on to attribute to him.

> By contrast, Lewis held that sentences are prior to grammars. While granting that we should not "discard" notions like phrasal meaning (relative to a population P), or the "fine structure of meaning in P of a sentence," he says that these notions "depend on our methods of evaluating grammars." … For Lewis, a grammar Γ is used by P if and only if Γ is a best grammar for a language £ that is used by P in virtue of a convention in P of "truthfulness and trust" in £. One might have thought that a "best" grammar for the alleged set £ would be one that best depicts the procedures acquired by members of P. But according to Lewis, "it makes sense to say that languages might be used by populations even if there were no internally represented grammars." He then makes an even more remarkable claim. "I can tentatively agree that £ is used by P if and only if everyone in P possesses an internal representation of a grammar for £, if that is offered as a scientific hypothesis. But I cannot accept it as any sort of analysis of "£ is used by P", since the analysandum clearly could be true although the analysans was false." Note the shift from Lewis's opening question—what is a language?—to a search for an analysis of what it is for a language to be used by a population. … This shift is interwoven with the insistence that languages are sets of sentences, and a willingness to accept the consequences for how grammars are related to non-sentential expressions.

What Pietroski describes here as a "shift" in Lewis's explanatory aims strikes me as a correct description of what Lewis was up to all along; or, at any rate, *should've* been.

Setting aside Lewis exegesis, we can turn again to Brandom, whose methodology is a good deal more perspicuous than his teacher's. As already noted, Brandom rejects the "set of sentences" conception of languages, tying social norms to practices that are fluid, socially distributed, and highly context-dependent. Nevertheless, he can accept Lewis's views (quoted in the passage immediately above) about grammars and about the status of subsentential expressions. Again, that's because *he* motivates the primacy of sentences (or "full thoughts") by reference to their roles in assertions, and hence in the norms governing the social practices in which assertion is a basic move. Such a practice need not be as articulated as ours, in respect of either content *or* syntax. And while Pietroski is prosecuting an empirical inquiry into *human psychology*, Brandom's is assaying the pragmatics of *assertion*, *inference*, and *assessment*.

Here is one final bit of textual evidence for my suggestion that Chomsky and Pietroski, on the one hand, and Lewis and Brandom, on the other, are simply talking at cross purposes. Pietroski writes:

> I think [Lewis's] ordering of priorities is misguided. Slangs are child-acquirable generative procedures that connect meanings with pronunciations in ways that allow for constrained homophony. So whatever meanings are, there are natural procedures that connect them with pronunciations in specific ways. Instead of adopting this promising starting point for an empirically informed discussion of languages and meanings, Lewis offered a series of stipulations. Many others followed suit. But that doesn't make it plausible that Slangs are sets. And if Slangs are I-languages in Chomsky' s sense, then we shouldn't ignore this fact when asking what meanings are. … [Lewis] adds that "the point is not to refrain from ever saying anything that depends on the evaluation of grammars. The point is to do so only when we must, and that is why I have concentrated on languages rather than grammars". But in reply to a worry that he is needlessly hypostatizing meanings, he says "There is no point in being a part-time nominalist. I am persuaded on independent grounds that I ought to believe in possible worlds and possible beings therein, and that I ought to believe in sets of things I believe in." So why be a part-time grammarist, given Chomsky's reasons for

> thinking that children acquire generative procedures? Quine's worries about the "indeterminacy" of meaning are not far away. But while Lewis speaks of evaluating grammars, he does not engage with Chomsky's notion of an evaluation metric, or the correlative notion of "explanatory adequacy".

I suspect that the reason Lewis would give for being what Pietroski disparagingly calls a "part-time grammarist" is that he is—as he says repeatedly—*not* concerned with specifically *human* languages, but with language *as a general phenomenon*. This, in any event, is the line that Brandom takes. And while Pietroski's arguments in favor of his semantic proposal are compelling in the context of *his* particular brand of inquiry, it's difficult to make sense of his idea that a different "ordering of priorities" can be "misguided". At worst, an ordering of priorities—i.e., the adoption of some concrete set of methodological maxims, descriptive aims, and explanatory ambitions—can fail to be illuminating about the domain that it carves out for itself. But it's hard to see how this charge can be credibly leveled against Brandom's inquiry, the results of which many philosophers—the present author included—find deeply revelatory of communicative linguistic practices.

§4.1.3 Public languages, dialects, and I-languages

Another context in which the notion of E-language has been invoked is to account for how folk terms like 'Italian' and 'Swahili' manage to pick out something in the world. Pietroski takes a dim view of this theoretical aim.

> we can describe many I-languages as English dialects (or idiolects), without English being any particular language. Prima facie, there are many ways to be a speaker of English: American, British, and Canadian ways; young child ways, adult scientist ways; etc. Being a speaker of English seems to be a *multiply realizable* property whose instances are similar in ways that matter for certain practical purposes. We can use 'English' to group certain I-languages together, perhaps in terms of paradigmatic examples, an intransitive notion of mutual intelligibility—think of Brooklyn and Glasgow—and historically rooted dimensions of similarity. There need not be an English language that each speaker of English has imperfectly acquired; cp. Dummett. We can use 'Norwegian' similarly, and classify both Norwegian and English I-languages as Germanic, without supposing that Germanic is a language shared by speakers of Norwegian and English. Analogies between linguistic and biological taxonomy can be preserved, whatever their worth, by thinking of specific I-languages as the analogs of the individual animals that get taxonomized—with 'Human' as the most inclusive category, and 'Indo-European' indicating something like a phylum.

Turning to Brandom, we note once more that one thing he is definitely *not* trying to do is to taxonomize the natural languages of the human species. He's simply not in the business of reifying or hypostasizing the norms that govern specific discursive interactions, nor individuating the entities that allegedly answer to the names 'English' or 'Norwegian'. There really seems to be no *principled* reason why Brandom couldn't countenance Pietroski's eminently reasonable view on these matters.

Thus, we can confidently assert that, when Pietroski describes a hypothetical theorist who "grant[s] that children acquire I-languages, yet maintain[s] that Slangs are E-languages that connect pronunciations with extensions of idealized concepts," the theorist he is describing is *not* Brandom. If the latter were in the business of theorizing about Slangs, in particular, then he might indeed take up the hypothetical proposal that "each Slang is a social object that certain speakers acquire by internalizing a generative procedure and meeting some *further* conditions." But no part of his actual view hangs on whether "English", "British English", or "Germanic" name metaphysically real entities, let alone ones that are individuated extensionally. Nor, again, is his theory a descriptive psychological one. And it most certainly has no truck with extensions.

We can illustrate all this more clearly by considering a contrast that Pietroski draws between what he calls E-NGLISH (an E-language) and NGLISH (an I-language).

> One might describe E-NGLISH in terms of the strings that certain people *could* understand as sentences, and the meanings they *could* assign to those strings, given a certain dictionary and a suitably idealized sense of 'could'. But this presupposes that a competent speaker of NGLISH has an expression-generating procedure whose lexicon can be expanded. So we don't need to invoke E-NGLISH to say what NGLISH is.
>
> One can stipulate that if Amy, Brit, and Candice speak English, there is something that Amy, Brit, and Candice speak. But then the "thing spoken" may be a class of I-languages. One can stipulate that people share a language if and only if they can communicate linguistically. But then "shared languages" may not play any role in *explaining* linguistic communication. Sharing a language, in the stipulated sense, may be a matter of using similar I-languages in combination with other human capacities and shared assumptions about potential topics of conversation.

These, I maintain, are all claims that Brandom should be happy to accept. True, he holds that there are social norms governing the communicative exchanges of creatures who can understand and produce indefinitely novel constructions. But he should have no qualms with the claim that humans happen to do this via some subpersonal psychological apparatus. Plainly, his account must presuppose that there is at least one way of "doing the trick"—our own—though it leaves room for others.

Ultimately, as I've emphasized throughout this discussion, the *appearance* of friction is, here as elsewhere, rooted in a cluster of verbal disputes. Underlying these is the fact that Brandom's normative inferentialism, while sharing many homophonous pieces of jargon with Pietroski, is pitched at a *higher theoretical plane*, so to speak, than a cognitive theory of NGLISH. Thus, Brandom should grant that NGLISH—the psychological apparatus that a given speaker possesses—can be specified without any recourse to E-NGLISH, if such a thing can even exists in a sense that he would countenance.

Note that, if Brandom's inferentialism is on the right track, then the "shared assumptions about potential topics of conversation" that Pietroski mentions are *partly constitutive* of the discursive norms that enter into an illuminating pragmatic account of communication. Needless to say, these not be the *only* things that enter into such an account. At the level of phonological and morpho-syntactic processes, as well as the ability to access and assemble concepts, the computational—and, ultimately, neurocognitive—explanation will surely be couched in the terms that generativists recommend. These are arguably *subpersonal* processes, which, again, is not the theoretical "level" at which Brandom's account is pitched.[19] A substantive disagreement can only be maintained if both inquiries have a shared target; this is, once again, *not* such a case.

Still, I think it must be admitted that a large part of what makes a cognitive theory of NGLISH theoretically interesting is that it enters into a larger account of how a specific creature—the human animal—is able to track, follow, articulate, challenge, reject, and revise the norms of discourse that have contingently arisen in its social *milieu*. This is not the *only* source of theoretical interest, of course. The biologically-realized combinatorial principles that constitute

[19] I spell out what I mean by "subpersonal" in Pereplyotchik (2017: §7.3)

NGLISH are a marvel of neurocognitive information-processing, and should be studied by natural science as such, with the fascination that grips empirical linguists and neuroscientists alike. Nevertheless, the internal operations of this apparatus would be of no *adaptive* use to a creature if they made no contribution to the shaping of a broader class of norm-governed social activities. And, biology aside, the beauty of the mechanism is only *enriched*, not diminished, when we take into account the social interactions that it makes possible.

Famously, Chomsky (2016) rejects the idea that language is an adaptation specifically designed for social/communicative purposes. He entertains an alternative hypothesis to the effect that the *narrow faculty of language*—what Pietroski is here calling 'NGLISH'—was initially an aid to individual thought, making it possible for recursive structures in the mind to be composed, and thus "entertained". This hypothesis strikes me as pretty implausible, but even if it turns out to be true, Brandom's claims would still hold. That is, it would remain the case that the creature's newly-structured individual-level thoughts played a role in its profound re-shaping of the prevailing social norms. Had its thoughts or judgments—i.e., its normatively evaluable commitments to things being thus-and-so—not *somehow* become entwined with broader social practices, they would not thereby have been *commitments*, whatever else they might have been. For, lacking a social existence, there would have been no source of a normative check on *what* the creature had committed itself *to*. The conceptual contents of the internal states of a solitary creature would thus be underdetermined to such an extent that it may be more accurate, not to mention fruitful, to view them as *subpersonal* states—or, at any rate, as something *other* than *judgments*. But this reeks of the kind of terminological legislation that both Pietroski and Brandom repeatedly warn against, so I'll leave the matter there.

Putting aside terminology, what Brandom is concerned to articulate is the general structure of pragmatic social statuses—e.g., commitment and entitlements—and how the norms that govern those statuses can be used in theorizing about the semantics of linguistic expressions in any language. The project is "general" in the sense that it designed to be applicable to *any* case of linguistic communication, not just the human case. How the norms of communication that we find in humans today might have evolved in the distant past of our particular species is, once again, no part of Brandom's explanatory target. Clearly, they emerged somehow, and the empirical story is bound to be fascinating. But all Brandom really needs to get *his* project going is the very broad claim—empirical, though perhaps only by a courtesy—that *some* natural psychological mechanisms underlie—and thus *explain*, in the descriptive naturalistic sense—any given creature's facility with social norms.

One might suspect that there is nevertheless a substantive *metaphysical* dispute in the vicinity. Brandom posits public linguistic norms, whereas Pietroski has no truck with public entities of any kind. But matters aren't so clear. Consider how Pietroski characterizes two hypothetical ontological views about what NGLISH really is: "We can identify NGLISH with an I-language, or perhaps a class of I-languages that differ only in small ways that are irrelevant for the purposes at hand." But what the difference could there possibly be between positing an E-language and positing "a class of I-languages that differ only in small ways that are irrelevant for the purposes at hand"? What ontological payoff, that is, can there be for a theorist who insists on I-languages *and resemblance classes thereof*, as opposed to public languages—or, at any rate, theoretically useful public linguistic entities (TUPLEs)?

Pietroski hints that the issue may have to do with other metaphysical features of E-NGLISH and NGLISH, including their *modal* properties: Whereas "a Slang seems to have its composition principles *essentially*, ... E-NGLISH includes no composition principles; the set contains only string-meaning pairs, atomic and complex." The difference is that sets are individuated by their members, but "any initial list of atomic expressions can be updated." Pietroski's point, if I understand it correctly, is that the ongoing process of language change creates a moving target for semanticists like Lewis. As new lexical items emerge (or "go extinct") in the actual world, the set that such a theorist intends to pick out changes. Indeed, Pietroski quips that "[i]dentifying Slangs with sets of expressions is like identifying animals with sets of molecules, and insisting that growth be described as replacing one animal with another. Even if this metaphysics is coherent, it may not cohere with plausible biology and linguistics" (57).

It isn't clear, though, why this sort of consideration couldn't be pressed just as hard with regard to I-languages. Although Pietroski says that "a Slang seems to have its composition principles *essentially*," he plainly acknowledges that I-languages, conceived of as a psychological procedures, can *also* be updated. For instance, "communicative failures can lead children to modify their (still modifiable) procedures for connecting meanings with pronunciations, subject to constraints." Indeed, it seems to be an empirical hypothesis whether all of the elements of Pietroski's *own* cognitive/semantic proposal in *CM* come online in the child's I-language at the same time. Perhaps I-languages initially allow only instructions for combining monadic concepts, and only add (limited) dyadicity later in the maturational process. If the latter, then we might ask, "Is this a *new* I-language?" More importantly, how could we tell?

On the assumption that I-languages have their lexical entries and composition rules essentially, we can adapt Pietroski's animals/molecules analogy, seeing a child's Slang as a succession of distinct I-languages—one for each day, week, month, or year. This entails that a child's Slang over the course of a year can be a set of significantly different I-languages. Though it is doubtless amenable to empirical inquiry, the question, "How many I-languages per day?" seems metaphysically awkward, at best. Similarly, Pietroski holds that "dialectical variation … makes appeal to a single set of English expressions seem silly." Rightly so. But, again, the same considerations apply to the I-languages of speakers who are competent in many "dialects" or "languages" (as the benighted folk call them).

What are the individuation conditions on I-languages, in light of these considerations? In Pereplyotchik (2017: ch. 3), I argued that the answers to such questions are not much clearer in the case of I-languages than for the case of E-languages. I suspect that it will remain so for the duration of sustained inquiry in the decades to come.

§4.2 Philosophy has not *failed cognitive science*

In a recent paper, provocatively entitled "Why Philosophy Has Failed Cognitive Science," Brandom argues that analytic philosophy, exemplified in the work of Frege, has devoted a great deal of energy to clarifying the nature of logical and semantic notions, but that we've thus far failed to properly hand off the fruits of our heritage to researchers in cognitive science.[20] The present section is devoted to a survey of the claims that Brandom makes about this alleged failure. I'll argue that Pietroski's work provides a direct counterexample to several of these

[20] Recognizing that cognitive science is comprised of many fields, Brandom aims his criticism more directly at philosophers who work on topics in cognitive psychology, developmental psychology, animal psychology (esp. primatology), and artificial intelligence, *rather than* at those who study topics in neurophysiology, linguistics, perceptual psychology, learning theory, and memory. Admittedly, this is a strange way to cut up the terrain. In particular, for our purposes, it's not at all clear why philosophers of *linguistics* are not on Brandom's list of targets. But let's not dwell on this. If only for the sake of furthering our present inquiry, I'll include philosophers and language and linguistics in the list, making no exception for myself.

claims, but that Brandom is right to point out that many theorists, including Pietroski, hold commitments that Frege's insights should lead us to reject.

§4.2.1 Modes of inquiry, philosophical and scientific

Brandom recounts the way in which modern approaches to logic and semantics began with Frege's *Begriffsschrift*, which furnished us with a new logic and new ways of thinking about meaning. Russell then showed us how to apply these ideas more generally in philosophy. But, while the ideas that Frege and Russell developed about logic and semantics were quite *general* in their import, later theorists to attempted, with variable success, to apply those general ideas to the specific case of natural language—i.e., the system of representation that normal human children acquire. (I suspect this has to do our sheer familiarity with the only clear case of language use available—i.e., our own—coupled with the anthropocentrism that motivates any inquiry into language.) This had the effect of blurring the lines between a general *philosophical* theory of language, on the one hand, and an empirical linguistic inquiry into the special case of human linguistic competence. That, Brandom maintains, is a mistake. On his view, the kind of inquiry that Pietroski is engaged in deals with the contingencies and the specifics of how humans acquired conceptual and linguistic abilities. Philosophy, by contrast, deals with the "normative" question of what *counts* as "doing the trick".

Although I've followed Brandom in putting the point this way through the discussion so far, I must now register that this is not, in my view, the best way of saying what I think Brandom intends to say here. At any rate, it's not, by my lights, the point that he *should* be making at this juncture in the dialectic. For, one might legitimately wonder how "What counts as doing the trick?" gets to be a *normative* question—whether in the case of language or of anything else—rather than a straightforward question of fact. Presumably, "What counts as being a horse?" is not a normative question, for the simple reason that horses are a natural type of object, studied as such by zoologists.

A theorist sympathetic to Brandom might reply that the notion "counts as" is normative because it's a matter of what competences and abilities it is *appropriate* to ascribe to a creature. But, here again, a parallel move can be made in the case of horses, vis-à-vis the properties that are *correctly* ascribed to *them* (notably, the property of *being a horse*). As a friendly amendment to Brandom, then, I will address this worry on his behalf by re-iterating and fleshing out the proposal that I floated earlier in the discussion, regarding "levels of analysis".

As I've noted, it seems to me that what Brandom (and probably Lewis) has on offer is a *high-level description* of a theoretically interesting kind of social practice—specifically, ("what counts as a") *language* game—with all of the impressive social, practical, and cognitive benefits that make this kind of practice worthy of careful study. Correspondingly, I suspect Pietroski's proposal has its home in an inquiry pitched at a lower level of analysis. To flesh the picture out further, it will be useful to quickly rehearse a central tenet of the mainstream approach to "levels of analysis" in contemporary philosophy of science.

A view that posits multiple levels of theoretical analysis, whether in biology, computer science, or in the sciences overall, is not *thereby* committed to any particular story about how theoretical progress at any one level can, should, or must constrain theorizing at any other. True, the early proponents of a "levels of science" picture also attempted—unsuccessfully, as it turns out—to secure a "unity of science" thesis. But later thinkers, notably Fodor (1975), generously disabused us of these lofty ambitions. What we know now is that theoretical pressure can and often does "go both ways", with higher and lower levels informing one another in equal measure, and with equal authority. Lower levels, as such, are no longer seen as having an inherent epistemic privilege.

Similarly, we can now appreciate the fact—poorly understood until fairly recently—that theories at different levels of inquiry are often to a large extent *independent variables*. Here's what I mean: a theorist who formulates a high-level analysis of some phenomenon typically assumes—often with good grounds—that the generalizations they discover at *that* level might be implemented in any number of ways by lower-level mechanisms. That's one half of the independence claim. The other half is best appreciated from the perspective of a theorist working at the (relatively) *lower* level of analysis. From *this* vantage point, the mechanisms, laws, generalizations, and/or principles that are discovered, however "abstract" they might seem, are assumed to be just one instance of an even *more* general phenomenon—a token of a potentially much larger type.

What I want to recommend is that we apply these general considerations from the philosophy of science to the concrete case of generative linguistics and normative inferentialism. Although it would be misleading to say that the subject matters that these two research programs seek to address are literally *orthogonal* to one another, the grain of truth in that bit of imagery is this: Brandom high-level account is, as such, indifferent to how lower-level mechanisms might operate in various token instances. Pietroski, on the other hand, is assaying the fine-structure of the *lower*-level mechanisms, but only in the special case of human languages. As such, while the results of his inquiry are *relevant* to Brandom's overall picture—indeed, they might pose devastating problems for Brandom (see below)—they function in practice *not* as substantive theoretical constraints, but as an account of a *very* special case (particularly to *us*!) of the kind of story that Brandom presupposes can be told for *any* creature to which his normative pragmatic account is applicable.

If this is on the right track, then how do we make sense of the fact that Brandom, like Lewis, explicitly appeals to data from natural language in motivating his analyses of phenomena that are, at least in principle, specific to *our* way of doing things? If the theory is not intended to be a contribution to a "merely parochial" inquiry about *us*, then why use examples from *our* language—indeed, almost exclusively from English, in particular—in constructing and developing it? There are two complementary ways of answering this question. The first, already mentioned, is to point out that Brandom draws our attention to features of human languages (in practice, just English) *not* for the purpose of displaying empirical data that his account can explain, but, rather, to illustrate aspects of language that he believes have pragmatic or semantic analogues in languages *beyond* the human case. (By analogy, think of the Chomsky hierarchy.)

The second prong of the reply consists of highlighting the fact that Brandom has devoted much time and effort to arguing—ultimately persuasively, in my view—that many of the linguistic devices he treats in his work are actually universal and essential features of language as such. Moreover, as many generative linguists have pointed out in discussions of Universal Grammar, language universals need not be *categorical*; they can, instead, take a *conditional* form, e.g., "any language that has feature F will also have property P," or "if a language can express content C, then it can also express content C*." Brandom (2008) works out a detailed typology of such relationships between logically possible languages, including those that differ either in respect of their *general* expressive power or in respect of more specific semantic devices, e.g., deixis. He takes this to be a pragmatic-*cum*-semantic version of Chomsky's famous analysis of the *syntactic* hierarchies of expressive power.

The view that Brandom promotes throughout his discussions of this topic is that traditional philosophers of language, starting with Quine, directed their efforts at analyzing linguistic constructions that, by and large, shed light on quite general semantic phenomena—i.e., ones that we can hope to one day discover in other species (terrestrial or otherwise), or to build into our intelligent robots. Although such linguistic devices might seem, from the perspective of a modern-day linguist, to comprise a rather motley collection—why propositional attitude reports but not, say, ergative verbs?—the tie that binds them, according to Brandom, is one that we can best appreciate from the vantage point of a (high-level) normative inquiry into general pragmatics. The linguistic phenomena that Quine and others identified early on as being particularly germane to philosophy all have this in common: for each of them, there are good reasons to think that it's *not* just something we *happen* to find in distinctively human languages, but something that tells us about what a language *is*, irrespective of which creatures happen to use it or what subpersonal mechanisms they deploy in doing so.

§4.2.2 Frege's insights

The lessons that Brandom believes philosophers have failed to pass on to their colleagues in the sciences pertain to four key distinctions, all due to Frege, between (i) labeling and describing, (ii) freestanding and embedded content, (iii) content and force, and finally (iv) simple vs. complex predicates.

The last of these, Brandom argues, opens up a semantic hierarchy that is no less important for cognitive scientists to be familiar with than the syntactic hierarchy that bears Chomsky's name. Taking this hierarchy into account in the context of empirical theorizing would help, he claims, to characterize the phylogenetic and ontogenetic development of linguistic and conceptual capacities, upward through what Brandom thinks of as "grades of conceptual content", including the propositional variety, the quantificational refinement, and ultimately the *relational* contents that Frege taught us to recognize.

We've seen that Pietroski has a great deal to say about this. Indeed, the Fregean considerations that he surveys in the service of an avowedly naturalistic theory in cognitive science are precisely those that Brandom recommends to our attention (and *then* some). For Brandom, Frege's insight is that there are patterns in sentences that *cannot* be modeled as mere part-whole relations. For instance, although there is no expression that appears in "Herbie admires Jessica" and "Jessica admires Herbie" that doesn't *also* appear in "Herbie admires Herbie", the latter sentence exhibits an inferential pattern different from the other two—the patter that we gesture at by employing notational distinctions between, e.g., admire(x,y) and admire(x,x), or by making explicit their inferential proprieties by embedding them inside of conditionals, as so:

(2) If someone admires anyone, then someone admires someone. (true)
(3) If someone admires anyone, then someone admires *themselves*. (false)

Thus, admire(x,x) expresses a kind of predicate that is not a *part* of a sentence, but an *aspect* of it—which we can recognize as an "inferential pattern" and *model* as an equivalence class of sentences. Frege's device of function-application is a way of capturing this idea. Functions are not, in general *parts* of their outputs. (The function *capitol-of(x)* yields Kiev when applied to Ukraine, but neither *capitol-of(x)* nor Ukraine are parts of Kiev.) This is why sentential connectives can be modeled with Venn diagrams, but complex predicates cannot. Even the simplest mathematics uses complex predicates—e.g., *natural number* or successor(x, y)—and

Frege showed that, once you can build complex predicates, you can keep building endlessly more, in the manner we ran across in our discussion of Lewis's type-theoretic semantics.

As we've seen, Pietroski warns against taking for granted a creature's ability to construct concepts of unbounded adicities. But the warning is intended to apply *only* when doing natural-language semantics. For *other* purposes, Pietroski agrees that Frege's insights are of foundational and lasting importance. Moreover, the hypothesis that he develops posits thoughts that admit of a Fregean semantic treatment (perhaps even a truth-conditional one), but it requires these to first be converted, via Frege's process of concept invention, into the kinds of thoughts that are "legible", so to speak, to the human FL. While it's not clear what *independent* empirical evidence Pietroski might offer for positing psychological mechanisms that facilitate such a translation—I am aware of no obvious analogue in the case of other perceptual modules—what *is* clear is this: Brandom's contentions regarding Frege's distinction (iv), between simple and complex predicates, are rendered moot by the very *existence* of Pietroski's work, which presents an up-and-running empirical inquiry that is deeply informed by Frege's core contributions.

Matters are much less clear with regard to the other three distinctions that Frege was at pains to draw. Let's turn now to his distinction between labeling and describing.

§4.2.3 Sentences, predicates, and classification

Brandom points out that old-school scholastic accounts of thought were rooted in a classificatory account of concepts—a relic of Aristotelian "forms". The medievals noticed that, once you have singular terms and classifications, you can build *up* to an account of truth, and then analyze good inference in terms of truth-preservation. Pietroski unabashedly endorses this strategy—in particular, the Aristotelian focus on classificatory concepts, which are central to his predicativism about New Mentalese.

This raises the question: What exactly *is* classification? How does a predicate get to perform its semantic function? Here is Pietroski's answer:

> …intuitively, a predicate classifies things, into those that meet a certain condition (e.g., being a rabbit) and those that do not. Anything that meets the condition satisfies the predicate, which applies to anything that meets the condition. We can invent perceptible predicates. Though for now, let's focus on predicative concepts, like instances of RABBIT. I assume that many animals have such mental predicates. … [A] predicate may apply to each of [several] things, or to nothing. But these are just special cases of classifying. …even if logically ideal predication is relational as opposed to classificatory, there seems to be a psychological distinction between relational and classificatory concepts, even if we speak of monadic/dyadic/*n*-adic predicates.

What I see, both here and throughout *CM*, are inter-definitions of semantic notions like "applies to," "classifies," "satisfies," and "meets conditions." Although Pietroski has made it clear that he is *not* trying to "break out of the intentional circle," to use Quine's phrase, the account he provides does not, to my mind, do much to illuminate the phenomenon in question. Having more labels for it does allow us to conjure different clusters of theoretical intuitions. But none of these seems definite enough to make progress with.

Turn, then, to Brandom's answer, which has the virtue of laying out substantive proposals and refining them, arriving ultimately at one that meets various important desiderata. Like Pietroski, Brandom maintains that "classifying" is *not* the obtaining of some (super)natural relation between a concept and (a portion of) the actual world—let alone non-actual *possible* worlds. On his view, there are, instead, *acts* of classification—e.g., asserting "That's a rabbit," or tokening the corresponding perceptual thought ("LO, IT RABBITETH!"). We've already seen the details of Brandom's account of assertion, as well as his (subordinate) account of classification. Let's now approach the latter from a different direction, this time contrasting Brandom's view with extant rivals.

If asked, straight-out, "What is classification?," the knee-jerk response that most philosophers would offer is that classification is a matter of *differential responsiveness*. This is a start, but it leaves wide open the question of what vocabulary we're permitted to use in describing the objects, properties, and events to which a physical system might be differentially responsive. If we give ourselves free reign, then the notion becomes too cheap to do serious work; differentially responding to Italian and French operas would count as classifying them, regardless of how the trick was done. But, of course, one wants to know *how* that sort of thing happens, not just *that* it does. Unfortunately, pursuing the answer to this explanatory question by *restricting* our vocabulary to only naturalistically respectable terms quickly lands us with panpsychism—a bridge *much* too far. For, as Brandom points out, even a chunk or iron differentially responds to varying amounts of oxygen in its surroundings, e.g., by rusting.

Equally vacuous is the (unqualified) suggestion that we acquire predicative concepts, and hence classificatory powers, by performing a process of "abstraction" from either the intrinsic qualities of states of sentient awareness—as Hume, Russell, and Carnap all held at various points in their otherwise distinguished careers—or from the raw information supplied by sensory mechanisms, as naturalist like Neurath might have it. Without a detailed and well-motivated account of the operation of "abstraction", the acquisition of classificatory concepts has been *labeled* once more, but remains stubbornly *unexplained*.

To their credit, naturalists like Fodor and Dretske attempted to meet the problem head-on. Information-carrying states count as classificatory concepts, they argued, when they're embedded in suitably complex systems—ones that reliably keep track of their environment, learn, and behave flexibly, perhaps on account of their history of natural selection and innate resources. Burge (2010) adds to this list the requirement that the reliable tracking abilities must have the shape of perceptual *constancies*, not mere sensory registrations.

Brandom maintains that no such project can work, even in principle, precisely because it ignores Frege's conceptual distinction between mere *labeling* and full-blown *describing*. A case of labeling is one in which items are differentially sorted, but only extensionally, such that no specific inferential consequences can be drawn from the presence of the label. A magic wand might tell us that doorknobs, pet fish, and crumpled shirts are all and only the items that share the magic feature, F. But without knowing what F is, in intensional terms, we have no idea what, if anything, *follows from* the application of the label 'F'—what, in the fictional scenario, is semantically achieved by the activation of the wand. In order for this (or any other) physical signal to become more than a mere label, it must be inferentially articulated, in the sense that there have to be things that *follow from* something's being F, as well as things that can have an instance of 'F' in *their* inferential consequences.[21]

[21] Following Dummett's counsel, Brandom urges that we take into account *both* the circumstances *and* the consequences of applying a concept. For some nonsynonymous propositions, the antecedent circumstances coincide, but the inferential *consequences* serve to distinguish their contents. For instance, consider the contrast between "I will one day write a book about anarchism" and "I *foresee* that I will one day write a book about anarchism." The inferential antecedents ("circumstances") of these two claims might be the same, but the

One of Frege's key lessons, then, is that inferential significance is central to conceptual content. Some concepts have *only* inferential conditions of application, not perceptual ones—either contingently, as with GENE, or necessarily, as with POLYNOMIAL or FRICTIONLESS PLANE. One can, of course, call things "concepts" even when they meet less stringent conditions. But, in that case, one should be sure to note the difference between differential responsiveness and inferential articulation. Moreover, these points hold irrespective of whether a differential-response capacity is innate or learned, and they apply just as much Boolean *compounds* of more basic units of differential responsiveness—i.e., *compound* labels.

While it's impossible to credit Brandom's claim that philosophers like Pietroski have taken insufficient notice of Frege's foundational insights, there *is* something to be said, I think, for his criticism on this particular point. As we'll see below, Pietroski's views on classification don't seem to respect the distinction—whether it be Frege's or Brandom's—between labeling and describing. This has downstream consequences for Pietroski's view that really *do* seem to be out of step with Brandom's theoretical commitments.

The disagreement about classification is joined when Brandom asserts that *thinking about something*, as in "We're still thinking about his tax returns," is a matter of tokening *complete thoughts*—i.e., intentional states that can be expressed by linguistically competent creatures only in complete speech acts, which requires producing *complete sentences* (if only in the paradigmatic case). Pietroski, by contrast, follows the peculiar philosophical convention of using the phrase "thinking about" to denote a punctate event of conceptual classification. While he agrees with Brandom that having a thought requires tokening a "sentential concept", he also maintains that *all* concepts are "ways of thinking about things."

This is where Brandom would disagree, on account of his commitment to the effect that *sub*sentential concepts are *not* complete thoughts. According to him, tokening such a concept cannot *by itself* constitute "thinking about something". To do *that*, subsentential concepts must (in some way) participate in a *sentential* one. So while sentential concepts are correctly described as "ways of thinking about things," Brandom follows Frege in viewing *sub*sentential concepts as *aspects of* such ways. Thus, whereas Pietroski claims that "hearing 'Bessie' can… activate the denoter BESSIE, thereby leading [one] to think about Bessie in a singular way" (108), Brandom would deny that activating the denoting concept BESSIE can *alone* constitute thinking about Bessie—in *any* way—even *once*. This point about "denoters" applies also, *mutatis mutandis*, to predicative concepts.

Pietroski can, of course, *stipulate* that thinking about things doesn't require tokening complete thoughts. But it's difficult to see what could motivate such a move. Relying on brute introspection, one might fancy that singular reference has taken place with only one subsentential concept in play—e.g., "I'm quite certain that I was just thinking of tofu; not anything about it, specifically; just… *tofu*." However, such introspective judgments are known to

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inferential consequences are different. This point applies even to observational concepts—e.g., MOVING or MOTION. A motion detector or a well-trained parrot that reliably emits the sound /*Moving*/ when there is, in fact, movement afoot (and not otherwise) does not thereby have the concepts in question. For although the circumstances of application are right, there are no inferential consequences to speak of in these cases. Brandom also makes the helpful observation that operators can serve to distinguish concepts that share both circumstances *and* consequences of application. For instance, the concepts HERE and WHERE-I-AM are shown to be distinct when interacting with the temporal operator 'always': "It's nice here/where I am" vs. "It is *always* nice here/where I am."

be an extremely unreliable source of data, whether performed by naïve speakers or by theoreticians.[22] One might, more plausibly, appeal to the theory of *perception* developed by Burge (2010), according to which perceptual awareness involves the application of only *sub*sentential concepts, modeled on noun phrases. But this won't do, either. For, if judgments and classifications are all "sentence-sized", as Brandom argues, then even the *perceptual* mental attitude of *noticing* can't properly be treated as a case of applying just *one* classificatory concept. Noticing rabbits involves *judging that* there are rabbits in the relevant spatiotemporal vicinity, and making such judgments requires deploying concepts *other* than the classificatory predicate, RABBIT(_)—e.g., the concept HERE(_).

With all this in mind, I think we should side with Brandom in saying that subsentential concepts *play a role* in acts of classification, where the latter are construed as either as public assertions or as inner endorsements of judgable contents. I see no reason to assume that tokening a subsentential concept is sufficient to carry off an act of classification. Nor is it obvious that classifying is a function of *all* concept application, as Pietroski believes. Does wondering whether Bessie exists really require *classifying* her? The latter question brings us face-to-face with the Fregean distinction between force and content, to which we now turn.

§4.2.4 Force and content

Brandom draws our attention to an ambiguity that was long ago pointed out by Wilfrid Sellars—the so-called "-ing/-ed" ambiguity—which allows us to use words like "claim" and "thought" polysemously to describe speech acts and propositional attitudes in respect of their intentional contents, on the one hand, and in respect of their illocutionary force or "mental attitude type", on the other. With regard to the latter, Stephen Schiffer has popularized the imagery of different "boxes" in the mind—one that corresponds to the functional role of beliefs, another to that of desires, a third one for intentions, and so on. Pietroski likewise notes the distinction in the following passage from *CM*.

> One needs to be careful with the terminology, since words like 'thought' and 'concept' are polysemous with regard to symbols and contents; 'thought' and 'judgment' are also like 'assertion', which can be used to describe certain events that can be characterized in terms of contents. In speaking of a thought that Sadie is a horse, one might be talking about a mental episode, a mental sentence, or a content shared by various sentences.

Brandom goes on to argue that this distinction is not only useful for theorists, but that it also marks *a distinct level of conceptual sophistication*. Creatures who can tell the difference between the act of *asserting* and the content of what's *asserted* can be said to be aware, at least *implicitly*, of the force/content distinction. To make this awareness *explicit*, a creature can embed a sentence inside of a conditional, thereby stripping it of its force.[23]

Now, on the assumption that classification *is*, in fact, a kind of illocutionary force, Brandom concludes that 'If F*a* then G*a*' *cannot*, in point of fact, be used to classify *a* as F, despite invoking

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[22] Distinct methodological troubles plague both of these two options, but they all strike me as insuperable and not worth discussing here.

[23] Brandom illustrates how conditionals can be used to distinguish those inferential consequences that derive from the *content* of what's said from those that derive from its *force*. Witness, for instance, the strikingly different inferential consequences of the sentences "p" and "I believe that p" when embedded as antecedents in conditionals: "If p then p" is obviously true for all values of 'p', but "If I believe that p, then p" is not foolishly arrogant for a mere mortal to assert, but also disastrously false in all known cases.

both ‘*a*’ and ‘*F*’.[24] This is another place where his views on the nature of classification come into conflict with Pietroski’s. And, here again, I can think of no plausible way around it.

One might suggest, on Pietroski’s behalf, that we seem to be able to simply *entertain a notion*—e.g., to contemplate “justice” or “the possibility of pigs one day flying”—without thereby committing ourselves to anything at all. This, Brandom points out, goes back to Descartes’s view that one can first “entertain” an idea/proposition and then, by *an act of mental will*, either endorse or deny it, yielding either a committal judgment or a positive doubt. Pietroski’s picture of concept-assembly likewise points in this direction. On that model, the process of assembly eventuates in the construction of a “polarized sentential concept”, which is then shipped off to central cognition for endorsement, rejection, or further contemplation.

But this idea is at odds with Kant’s equally compelling observation that concepts have contents *only* in virtue of their role in judgment. Pushing still further, Frege argued that entertaining propositions is a late-coming ability that involves a thinker embedding a proposition into the antecedent slot of a conditional—as in the following soliloquy: “What if *p*? Well, if *p* were the case, then *q* would also; but *that* would mean that neither *r* nor *s*…”. If Frege’s proposal is correct, then the ability to “entertain an idea” piggy-backs on two *prior* abilities—viz., to assert conditionals, and then to perform inferences that take *them* as premises or conclusions (e.g., hypothetical syllogisms).

Now, Pietroski agrees that the mental act of endorsement results in a committal *judgment*, which both he and Brandom take to be subject to normative evaluation—i.e., assessments of correctness, warrant, rational propriety and the like. But it’s not clear how Pietroski’s ⇑/⇓ operators for assembling polarized sentential concepts facilitate this act of endorsement. More generally, Pietroski’s proposal seems to have little to offer in the way of a subpersonal about how *any* kind of force/attitude is superadded, so to speak, to polarized concepts, after the *Begriffsplans* get done assembling them.

§4.3. Predicates and singular terms

We turn now to our very final topic, which concerns a foundational disagreement between Brandom and Pietroski on the nature of singular and predicative concepts. Recall that Pietroski’s semantics for natural language is resolutely *predicativist*, in the sense that it recognizes no analogue of type-<e> expressions—intuitively, singular terms—i.e., no instruction for fetching singular concepts. Recall as well that he *does* countenance the presence of such concepts in the human mind and that he recognizes the useful cognitive roles that such concepts play in thinking/reasoning. But this kind of cognition is couched in Olde Mentalese—the phylogenetically ancient representational format in which pre-linguistic thought was conducted, and which Pietroski thinks we still employ today, outside of our language use.

Brandom develops a powerful argument to the effect that *any* language that fails to draw a distinction between predicates and singular terms is in principle barred from introducing basic logical operators—including both negation and the conditional. If this argument is successful, it would have no effect at all on Pietroski’s claims about Olde Mentalese, which happily draws that distinction. But it would seem to present a rather major problem for Pietroski’s main proposal about natural-language semantics, which has predicativism as one of its core commitments. So

[24] Likewise, he warns against conflating denial and supposition—two kinds of force—with negation and conditionalization, which are semantic functions that directly participate in the content

it behooves us, in surveying the points of discord between them, to focus on this foundational case, using it to draw out related points of contention about *syntax*.

§4.3. Brandom's argument

"What are subsentential expressions?" and "Why are there any?" These are the two questions that Brandom raises in an essay of the same title (2001: ch. 4). In §1, we glimpsed the overall shape of his answer. Here, we'll reiterate the main points and look at some of the details. The reason for doing so is that this is the last—and arguably most challenging—of the issues that divide Brandom's normative inferentialism from the overall generative enterprise.

In the reconciliatory spirit of my overall project, I'll propose a possible strategy for ameliorating the dispute. But I should concede from the outset that this appears to be a particularly stubborn issue. This is frustrating, as the issue obviously cuts pretty deep. Having laid out the details of Brandom's difficult argument, I'll settle, in the end, for merely having *raised* the question—one that hasn't been discussed, to my knowledge, anywhere else in the literature—of how generative grammar might be (in)compatible with Brandom's substitutional approach to syntax.

§4.3.1 Details and a proof

As noted earlier, Brandom agrees with Pietroski that discerning subsentential expressions is what makes it possible for us, both as theorists and as language users, to "project" proprieties governing the use of *novel* sentences. Once we've done this, we can then recombine subsentential items into new expressions, with meanings/contents that were previously inexpressible. Brandom recommends using the notion of substitution for this purpose, adapting Frege's insight that discerning meaningful subsentential expressions is a matter of treating sentences as *substitutional variants* of one another.

In spelling out the syntactic side of this technical notion, Brandom begins by identifying three "substitution-structural roles". These include the role of being an expression that is (i) substituted *in*, (ii) substituted *for*, and (iii) a substitutional *frame*. For instance, "David admires Herbie," is substituted *in* to yield a *substitutional variant*, such as "Herbie admires Herbie," where the expression 'David' has been substituted *for*. The residual substitutional *frame* is what's is common to the two substitutional variants—schematically, "*x* admired Herbie."

On the semantic side, a substitutional variant of a sentence will be defined in terms of the inferences that it enters into, as a premise or a conclusion. In keeping with his inferentialist project, Brandom develops the idea that the meaning of a subsentential expression consists in the materially correct *substitution inferences* involving that expression—i.e., inferences in which the conclusion is a substitutional variant of one of the premises. Thus, 'Herbie' has the meaning that it does partly in virtue of its role in a vast range of materially good inferences, including the single-premise inference from "Herbie barked" to "My dog barked".

With this in mind, Brandom notes that substitution inferences come in two flavors: symmetric and asymmetric. The above inference, from "Herbie barked" to "My dog barked", is symmetric, in the sense that it's materially good in *either* direction. Plainly, this trades on the identity between Herbie and my dog. This is more grist for Brandom's logical expressivist mill. He captures this observation by pointing out that identity is the logical notion that we use to *express*—i.e., make explicit—the *substitutional commitments* that are central to our notion of singular terms (and, relatedly, of the items they purport to denote). Contrast this with the inference from “Herbie runs” to “Herbie moves”, which is materially good in only one direction, not in the other. That’s because ‘runs’ is materially stronger, in respect of inferential consequences, than ‘move’; the former licenses all of the inferences that the latter does, and then some. The distinction between symmetric and asymmetric inferential proprieties governing substitution inferences is, as we’ll now see, the *central* aspect of Brandom’s distinction between predicates and singular terms. Let’s turn finally to his definitions of these two notions.

Each of the two definitions has a syntactic component and a semantic component. On the syntactic side, Brandom says that singular terms invariably play the substitution-structural roles of being substituted *for* (as well as *in*), whereas predicates invariably play the role of substitutional *frames*. On the semantic side, he points out that the substitution of singular terms is *always* governed by symmetric inferential proprieties, whereas predicates are *necessarily* governed by *at least some a*symmetric ones. For instance, ‘Herbie’ is a singular term partly in virtue of the fact that, if the substitution inference from “Herbie barked” to “My dog barked” is materially good, then so is its converse. Crucially, the same does *not* hold for the substitution inference from “Herbie runs” to “Herbie moves”, where the substitution of *predicates* is in play. That’s, again, because ‘runs’ is inferentially stronger than ‘moves’. This is an instance of something that Brandom goes on to argue is *constitutive* of predicates as a class—viz., that they *necessarily* enter into at least *some asymmetric* substitution-inferential relations with other predicates in the language.

Thus far, Brandom has supplied an answer only to his first question: what are singular terms (and, by extension, predicates)? To summarize, the answer is that singular terms play the *syntactic* roles of substituted *for*s and substituted *in*s, and the *semantic* role of entering *solely* into *symmetric* substitutional inferences. Predicates, by contrast, play the syntactic role of substitutional *frames* that *necessarily* enter into at least *some asymmetric* substitutional relations.

To ask why there are singular terms, then, is to ask the following question: Why do the syntactic and semantic substitutional roles line up as they do? This way of setting up the question allows us to generate a taxonomy of the logical possibilities, in terms of two binary parameters—syntax and semantics. We can thus imagine languages that instantiate the following four permutations.

i) Substituted *for* is *symmetric* and substitutional *frame* is *symmetric*.
ii) Substituted *for* is *asymmetric* and substitutional *frame* is *symmetric*.
iii) Substituted *for* is *asymmetric* and substitutional *frame* is *asymmetric*.
iv) Substituted *for* is *symmetric* and substitutional *frame* is *asymmetric*.

The option that’s *actually* instantiated by singular terms and predicates is (iv). The question then becomes: What's “wrong” with the other options?

What rules out option (i), according to Brandom, is that, many of the substitution inferences that are to be codified and projected at the level of sentences by discerning subsentential expressions are *asymmetric*. No weakening inferences could be generated if *all* subsentential components were restricted *solely* to symmetric inferences. What the remaining options have in common is that they assign *asymmetric* inferential proprieties to expression-kinds that play the syntactic role of being substituted *for*. We can thus ask: what’s wrong with *that* combination? The answer to this question is where things become technically challenging. Readers who feel like skipping ahead to the next section can take with them only the upshot of the proof: If a language fits the model of options (ii) or (iii), then it does not permit the introduction of conditional contents (contrary to fact, in our own case).

Brandom invites us to consider the generalizations that permit expressions with subsentential contents to determine the proprieties of a productive and indefinitely flexible class of novel combinations. Asserting that Herbie is (identical with) my dog commits me to the propriety of *all* inferences of the form P(Herbie) → P(my dog). Similarly, for predicates; asserting that anything that runs thereby moves commits one to the propriety of all inferences of the form Runs(x) → Moves(x). This is why, when such content-constitutive and potentially asymmetric substitutional commitments made explicit, they take the form of *quantified conditionals*—another feather in the logical expressivist's cap.

Here, then, is Brandom's proof that options (ii) and (iii) in effect rob a language of its most basic logical notions—even ones as simple as negation and the conditional.

The pattern corresponding to the hypothetical *asymmetric* significance of "substituted *for*s" would replace identity claims with *inequalities*. Let "$t > t^*$" mean that $P(t) \rightarrow P(t^*)$ is in general a good inference, but *not* every frame, P, will make the converse inference, $P(t^*) \rightarrow P(t)$, materially good. Now, call a predicate Q an *inferential inverse* of a predicate P if, for all t and t^*, the following condition is satisfied.

Inferential Inverse $=_{\text{df}}$ if $P(t) \rightarrow P(t^*)$ holds, but $P(t') \rightarrow P(t)$ doesn't,
then $Q(t^*) \rightarrow Q(t)$ holds and $Q(t) \rightarrow Q(t^*)$ doesn't

Thus, to answer the question of what's "wrong" with options (ii) and (iii), it suffices to show that if *every* sentential substitutional frame has an inverse, then there can be no *asymmetrically* significant substituted *for*s. The demonstration now proceeds by way of the following key observation.

Key observation: In any language containing the expressive resources of elementary sentential logic, every predicate has an inferential inverse. Conditional and negating locutions are inferentially inverting; e.g., inferentially weakening the antecedent of a conditional inferentially strengthens the conditional. Thus, if condition the antecedent of *Inferential Inverse* holds, the then the consequent can be shown to hold as well.

Proof: Let $Q\alpha$ be defined as $P\alpha \rightarrow r$. It follows immediately that $P(t^*) \rightarrow S(t^*)$ entails $P(t) \rightarrow S(t)$, but $P(t) \rightarrow S(t)$ does *not* entail $P(t') \rightarrow S(t')$.

What this argument shows, if it shows anything at all (see below), is that conditional locutions are inferentially inverting precisely *because* they play the indispensable expressive role of making inferential relations explicit. (*Mutatis mutandis* for negation and other logical operators.) If this is right, then we can conclude, as Brandom does, that *any* language able to muster the expressive resources required for introducing basic sentential connectives will also draw a distinction between singular terms and predicates (as defined), assuming it has any subsentential structure at all. Conversely, any language that *forgoes* the term/predicate distinction is thereby severely castrated in its expressive power—incapable in principle of introducing so much as a material conditional.

§4.3.2 Potential replies

The foregoing argument was developed in Brandom's was presented in its canonical form in *MIE*. (See also the later and more condensed treatment in Brandom, 2001). In the decades since then, many theorists have marshalled a variety of technical objections against his line of reasoning. Some of these are based on straightforward confusions and can thus be defused without much concern (see *MIE*, ch. 6). Others might be more troublesome. Whatever the case about that, I want to ask what bearing this argument *would* have on Pietroski's position if it *were* successful.

As noted above, the argument appears to present a serious problem for Pietroski's commitment to predicativism, at least in the case of natural-language expressions and New Mentalese. (We can breathe easy about expressions of Olde Mentalese; these are, mercifully, in the clear.) How might Pietroski reply to this challenge? Closer to home, if this dispute *can't* be resolved, does that spell doom for *my* larger reconciliation project in this essay? Sadly, the answers I can muster at present will be satisfying to few.

One possibility is cut things off at the root by rejecting Brandom's substitutional approach to both syntax *and* semantics. Indeed, this is most obvious route for Pietroski to pursue, given his claim that we can't simply take for granted a creature's ability to "slice out" terms from sentences, so as to use them in combinatorically constructing an infinite hierarchy of semantic types, *a la* Frege or Lewis. Such a project, Pietroski argues, stipulates from the outset far more than it explains in the end. Suppose that he's right about this. Does that mean that his empirical results—assuming for present purposes that that's what they are—have literally *contravened* Brandom's strategy? Put another way, if generative linguistics is the correct approach to natural language, then are we barred from using Brandom's "substitutional scalpel" to identify subsentential structure, distinguish between singular terms and predicates, and carry off the inferentialist project at the subsentential level? I do not think so. Or, at any rate, I'm not convinced.

One excuse for my wavering on this point is that the considerations Brandom uses are so general—i.e., so totally independent of other details of the languages to which it applies—that it's hard to see which of them Pietroski is really in a position to deny. True, substitutional syntax smells a little too much like the old-school "discovery procedures" and "immediate constituent grammars" of benighted pre-Chomskyans (see Fodor, Bever, and Garrett, 1974 for a blistering refutation). But the methodological *similarities*, in my view, cut no ice. Nominalist discovery procedures, were, for all their shortcomings, *empirical hypotheses* about human languages. Otherwise, they wouldn't even *get to be* rendered false by straightforwardly empirical arguments. By contrast, we've seen that Brandom's project, despite drawing on examples from English—again, *for illustrative purposes only*—is explicitly *not* pitched as an empirical inquiry into human language. So he can't be accused of attempting to resurrect *that* old idea.

Nor is it clear that Brandom's approach to the general project of delineating syntactic categories is incompatible with *further* elaborations by the kind of syntax that Chomsky supplies. In the only passage I've found where he mentions generative grammar and its transformation rules (Brandom, 1987), he makes precisely that suggestion:

> Recall … that Chomsky showed that one should not expect to generate the well-formed sentences of natural languages by concatenation, combination, or tree-structuring of any set of categories of this sort. To any such "phrase-structure grammar" will have to be added transformations of such combinatory structures. Categorial classifications are just

> the raw materials for grammar in this sense, and don't have anything to say about how one might proceed to the rest of the task of syntax once one has the categories. (165: fn. 2)

With this in mind, I'll end with the following throw-away speculation. If, in a remarkably distant possible world, Brandom *were* to go into the business of empirical theorizing, it's not clear to me that he would (or *should*) adopt anything other than generative grammar as the optimal theoretical framework within which to prosecute his inquiry. Certainly, he is well aware—how could he *not* be?!—of the theoretical need for Chomskyan grammars. On the rare occasion that he does mention these, he doesn't say anything that even hints at a disagreement. (Recall, "Chomsky *showed*…", my emphasis.) Moreover, his frequent invocations of the Chomsky hierarchy in discussions of computational procedures and the expressive power of various languages (e.g., Brandom, 2008: ch. 1) suggests no particular aversion to core generativist principles. To be sure, this isn't very much to go on. It doesn't even *begin* to show that Brandom's "substitutional syntax" is compatible with (any) generative grammar. But it's what I can offer at present, besides relentless optimism.

Conclusion

Attempting to integrate the theories developed by Brandom and Pietroski may strike some as an utterly futile project, analogous to grafting, say, a squirrel onto a cow. One thinks to oneself, "Perhaps it *can* be done, but… why?!" In the foregoing pages, I've argued that this view of the matter constitutes a failure to appreciate the live opportunities for a fruitful merger. Such a merger is, like any large one, a daunting gamble. But, it seems undeniable, from where I sit, that both Brandom and Pietroski have furnished significant insights into the nature of something called "language"—a phenomenon that we should firmly resist regarding as unitary.

That having been said, it seems only natural to suppose that *combining* the two theories will yield a richer *overall* picture than either theory can provide on its own. This sort of thing doesn't always work out, of course; not all teams of All-Stars are All-Star teams, after all. But even if the resulting view is not to one's liking, I find it frankly inconceivable that *some* such reconciliation project won't have to be effected *eventually*. Perhaps we aren't there yet; perhaps both generative linguistics and normative inferentialism must await more penetrating developments before their future descendants can be merged. (Or, again, maybe the AI people blow our minds with some new-fangled contraption next Tuesday. Who knows?) Whatever the case about that, I hope to have convinced the reader that there are, in fact, very few substantive *dis*agreements between the two approaches. What initially appear to be sharp contrasts turn out, on inspection, to be mostly benign differences of theoretical focus and explanatory ambition.

I'll close on a broadly sociological note. A mistaken commitment to the incompatibility of generative linguistics and normative inferentialism has had, I believe, negative consequences for both philosophy and linguistics. Specifically, there is, at present, little or no cross-talk between researchers working in these two traditions. Indeed, they seem to be as siloed off from one another as any two major research programs in "analytic" philosophy of language can be. If nothing else, by *partially* undermining the mistaken assumption of incompatibility, I hope to have gone at least some way toward rectifying the situation. My hope is that others will follow suit, attempting—surely with more skill than I—to forge still further connections between the two enterprises. Even if Pietroski and Brandom make for strange bedfellows, there is no question that they make for excellent guides. And, for better or worse, the terrain is largely uncharted. Let us press forward, then—as always, with optimism.

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